

École Polytechnique Fédérale de Lausanne

Thermal properties of MOFs from molecular dynamics

Master Thesis

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Abstract

Methane-loaded MOF-5 heat capacity is estimated by combining classical NPT molecular dynamics with a loaded-system harmonic quantum correction from PET-JAX/SADMOF Hessians. PET-MAD and PET-SOL calculations at 50, 100, and 150 methane molecules give exploratory room-temperature values of about 0.93, 1.07–1.08, and 1.16 J/g/K. Sampling errors and paired last-layer prediction-rigidity (LLPR) model uncertainties are propagated separately. Production drift, imaginary modes, incomplete relaxation, and sensitivity to uncertainty propagation prevent a claim of convergence. Variation across quenched minima remains unestimated. The empty-framework Hessian provides a separate reference.

1 Scientific motivation and scope

MOF-5 combines high-frequency bond vibrations with soft framework motion; methane adds internal vibrations and anharmonic motion within the pores. Classical MD samples the latter, while normal modes correct the excessive classical occupation of high-frequency vibrations. The conventional host cell contains 424 atoms, $C_{192}H_{96}O_{104}Zn_{32}$; 50, 100, and 150 methane molecules increase the totals to 674, 924, and 1174.

Figure 1 summarizes the calculation. All loaded terms use the same PET checkpoint, composition, and mass normalization. The aim is to test the hybrid approximation of Kapil *et al.* [21] on PET through convergence and validation, rather than matching their force-field results.

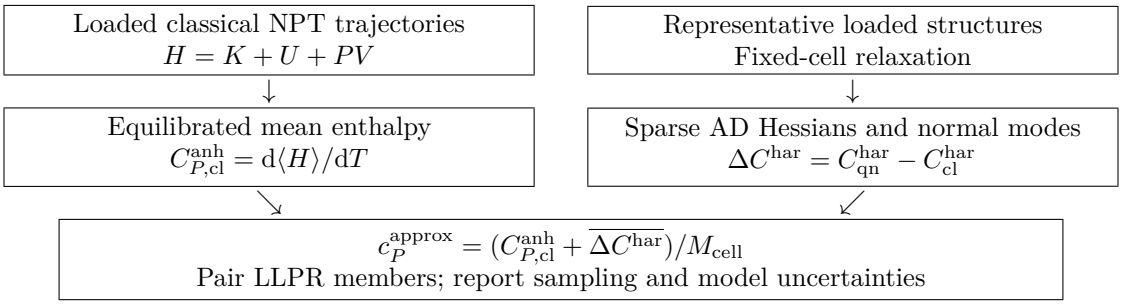


Figure 1: Loaded-system workflow. Accept harmonic references only after force and signed-spectrum checks. The empty-MOF Hessian is evaluated separately, without an empty-MOF MD campaign.

2 Theory

2.1 Quantum Mechanics

2.1.1 States, eigenstates, and observables

A quantum state is represented by a ket $|\Psi\rangle$ in a Hilbert space; its Hermitian conjugate is the bra $\langle\Psi|$. The inner product $\langle\Phi|\Psi\rangle$ is a complex probability amplitude, and a normalized state satisfies $\langle\Psi|\Psi\rangle = 1$. Observables are represented by Hermitian operators, denoted with a hat [1]. For an observable $\hat{A}$, an orthonormal eigenbasis satisfies

$$\hat{A}|a_n\rangle = a_n|a_n\rangle, \qquad \langle a_m|a_n\rangle = \delta_{mn}, \qquad \sum_n |a_n\rangle\langle a_n| = \hat{I}. \tag{1}$$

The eigenvalue a_n is a possible measurement result. Expanding $|\Psi\rangle = \sum_n c_n|a_n\rangle$ gives probabilities $|c_n|^2 = |\langle a_n|\Psi\rangle|^2$ and the expectation value

$$\langle A\rangle_\Psi = \langle\Psi|\hat{A}|\Psi\rangle = \sum_n |c_n|^2 a_n. \tag{2}$$

Here sums refer to a discrete spectrum; continuous labels require integrals. We denote the Hamiltonian operator by $\hat{E}$, consistently reserving H for enthalpy. For a time-independent Hamiltonian, its energy eigenstates and the evolution of a general state obey

$$\hat{E}|\Psi_n\rangle = E_n|\Psi_n\rangle, \qquad i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = \hat{E}|\Psi(t)\rangle. \tag{3}$$

These equations do not depend on a choice of basis.

2.1.2 The position representation

For the electron and nuclear positions collected in $\boldsymbol{r}$ and $\boldsymbol{R}$, the position eigenkets $|\boldsymbol{r},\boldsymbol{R}\rangle$ form a continuous basis. Spin labels are suppressed. The wavefunction is the projection of the abstract state onto this basis:

$$\Psi(\boldsymbol{r},\boldsymbol{R},t)=\langle \boldsymbol{r},\boldsymbol{R}|\Psi(t)\rangle, \qquad |\Psi(t)\rangle=\int \mathrm{d}\boldsymbol{r}\,\mathrm{d}\boldsymbol{R}\,|\boldsymbol{r},\boldsymbol{R}\rangle\Psi(\boldsymbol{r},\boldsymbol{R},t). \tag{4}$$

The integration covers all particle coordinates, and the bra has wavefunction $\langle\Psi|\boldsymbol{r},\boldsymbol{R}\rangle=\Psi^*(\boldsymbol{r},\boldsymbol{R})$. Position acts by multiplication and momentum by differentiation; for nucleus i,

$$\langle \boldsymbol{r},\boldsymbol{R}|\hat{\boldsymbol{R}}_i|\Psi\rangle=\boldsymbol{R}_i\Psi(\boldsymbol{r},\boldsymbol{R}), \qquad \langle \boldsymbol{r},\boldsymbol{R}|\hat{\boldsymbol{p}}_i|\Psi\rangle=-i\hbar\nabla_{\boldsymbol{R}_i}\Psi(\boldsymbol{r},\boldsymbol{R}). \tag{5}$$

For example, the pure-state expectation in Eq. (2) becomes

$$\langle A\rangle_\Psi=\int \mathrm{d}\boldsymbol{r}\,\mathrm{d}\boldsymbol{R}\,\Psi^*(\boldsymbol{r},\boldsymbol{R})\,A_{\mathrm{pos}}\Psi(\boldsymbol{r},\boldsymbol{R}), \tag{6}$$

where A_{pos} denotes the operator acting on wavefunctions in this basis. Neglecting relativistic effects, the electron–nuclear Hamiltonian is

$$\hat{E}_{\mathrm{tot}}=\hat{T}_{\mathrm{n}}+\hat{T}_{\mathrm{e}}+\hat{V}_{\mathrm{ee}}+\hat{V}_{\mathrm{en}}+\hat{V}_{\mathrm{nn}}. \tag{7}$$

Projecting its Schrödinger equation with $\langle\boldsymbol{r},\boldsymbol{R}|$ gives the position-space differential equation

$$i\hbar\frac{\partial\Psi}{\partial t}=\left[-\sum_i\frac{\hbar^2}{2m_i}\nabla^2_{\boldsymbol{R}_i}-\sum_a\frac{\hbar^2}{2m_{\mathrm{e}}}\nabla^2_{\boldsymbol{r}_a}+V_{\mathrm{ee}}+V_{\mathrm{en}}+V_{\mathrm{nn}}\right]\Psi. \tag{8}$$

The V terms here are coordinate-dependent Coulomb potentials. Thus a ket and its wavefunction describe the same state, while the differential operator is the position representation of the Hamiltonian. Solving the coupled many-electron and nuclear problem requires approximations; the first used here is the Born–Oppenheimer separation [2, 5].

2.1.3 The Born–Oppenheimer approximation

Because nuclei are much heavier than electrons, we approximate their motion on a single electronic ground-state potential-energy surface. At each fixed nuclear configuration $\boldsymbol{R}$, solve the electronic *operator* eigenvalue problem

$$\hat{E}_{\mathrm{el}}(\boldsymbol{R})=\hat{T}_{\mathrm{e}}+\hat{V}_{\mathrm{ee}}+\hat{V}_{\mathrm{en}}(\boldsymbol{R}), \tag{9}$$

$$\hat{E}_{\mathrm{el}}(\boldsymbol{R})|\psi_0(\boldsymbol{R})\rangle=\varepsilon_0(\boldsymbol{R})|\psi_0(\boldsymbol{R})\rangle. \tag{10}$$

Here $\boldsymbol{R}$ is a parameter, while $|\psi_0(\boldsymbol{R})\rangle$ is a state in the electronic Hilbert space. Its position representation is $\psi_0(\boldsymbol{r};\boldsymbol{R})=\langle\boldsymbol{r}|\psi_0(\boldsymbol{R})\rangle$. Projecting with $\langle\boldsymbol{r}|$ gives the equivalent equation

$$\left[-\sum_a\frac{\hbar^2}{2m_{\mathrm{e}}}\nabla^2_{\boldsymbol{r}_a}+V_{\mathrm{ee}}(\boldsymbol{r})+V_{\mathrm{en}}(\boldsymbol{r};\boldsymbol{R})\right]\psi_0(\boldsymbol{r};\boldsymbol{R})=\varepsilon_0(\boldsymbol{R})\psi_0(\boldsymbol{r};\boldsymbol{R}). \tag{11}$$

The semicolon distinguishes the electronic coordinates from the nuclear parameters. Adding nuclear repulsion defines the potential-energy surface $U(\boldsymbol{R})=\varepsilon_0(\boldsymbol{R})+V_{\mathrm{nn}}(\boldsymbol{R})$. Neglecting derivative couplings associated with the nuclear dependence of the electronic state gives the nuclear Hamiltonian

$$\hat{E}_{\mathrm{nuc}}=\sum_i\frac{|\hat{\boldsymbol{p}}_i|^2}{2m_i}+U(\hat{\boldsymbol{R}}). \tag{12}$$

Its eigenstates $|\chi_\nu\rangle$ and their nuclear wavefunctions $\chi_\nu(\boldsymbol{R})=\langle\boldsymbol{R}|\chi_\nu\rangle$ satisfy, respectively,

$$\hat{E}_{\mathrm{nuc}}|\chi_\nu\rangle=E_\nu|\chi_\nu\rangle, \tag{13}$$

$$\left[-\sum_i\frac{\hbar^2}{2m_i}\nabla^2_{\boldsymbol{R}_i}+U(\boldsymbol{R})\right]\chi_\nu(\boldsymbol{R})=E_\nu\chi_\nu(\boldsymbol{R}). \tag{14}$$

The corresponding approximation to the total wavefunction is a product *in the position representation*:

$$\Psi_\nu(\boldsymbol{r},\boldsymbol{R})\approx\psi_0(\boldsymbol{r};\boldsymbol{R})\chi_\nu(\boldsymbol{R}). \tag{15}$$

The electronic factor still depends on the nuclear configuration. Classical MD later replaces this quantum nuclear evolution by motion of positions and momenta on the same surface U.

2.2 Density Functional Theory

At fixed nuclear positions, density functional theory (DFT) expresses the electronic ground-state energy in terms of the electron density $n(\boldsymbol{r})$ rather than the many-electron wavefunction. This provides the Born–Oppenheimer potential-energy surface used to generate reference energies, forces, and stresses for machine-learned potentials.

2.2.1 The Hohenberg–Kohn theorems

For a fixed electron number N_e and external potential $v_\mathrm{ext}(\boldsymbol{r};\boldsymbol{R})$, the electronic Hamiltonian is

$$\hat{E}_\mathrm{el} = \hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee} + \sum_{a=1}^{N_\mathrm{e}} v_\mathrm{ext}(\hat{\boldsymbol{r}}_a;\boldsymbol{R}). \tag{16}$$

The Hohenberg–Kohn (HK) theorems establish the basis of ground-state DFT [6]:

1. The ground-state density $n_0(\boldsymbol{r})$ determines the external potential up to an additive constant, and hence the electronic Hamiltonian and ground-state properties.
2. The ground-state energy is the minimum of the density functional $E_v[n]$, attained at n_0, over admissible densities with $\int n(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} = N_\mathrm{e}$.

Writing the universal functional $F[n]$ in constrained-search form makes this variational principle explicit [5]:

$$F[n] = \min_{\psi\to n} \langle\psi|\hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee}|\psi\rangle, \tag{17}$$

$$E_v[n] = F[n] + \int v_\mathrm{ext}(\boldsymbol{r};\boldsymbol{R})n(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r}, \qquad E_0 = \min_{n\ \text{admissible}} E_v[n]. \tag{18}$$

HK therefore turns the electronic problem into a density minimization, but does not supply a practical expression for $F[n]$.

2.2.2 From HK to the Kohn–Sham equations

Kohn–Sham (KS) theory introduces an auxiliary system of noninteracting electrons with the same density [7]. Its orthonormal orbitals $|\phi_a\rangle$, written as $\phi_a(\boldsymbol{r}) = \langle\boldsymbol{r}|\phi_a\rangle$ in the one-electron position basis, and occupations f_a give

$$n(\boldsymbol{r}) = \sum_a f_a|\phi_a(\boldsymbol{r})|^2, \qquad \sum_a f_a = N_\mathrm{e}. \tag{19}$$

The occupations include spin multiplicity; for a periodic solid the sum also includes band indices and wavevector weights. The unknown HK functional is decomposed as

$$F[n] = T_\mathrm{s}[n] + E_\mathrm{H}[n] + E_\mathrm{xc}[n], \tag{20}$$

$$T_\mathrm{s}[n] = \sum_a f_a \int \phi_a^*(\boldsymbol{r}) \left(-\frac{\hbar^2}{2m_\mathrm{e}}\nabla^2\right) \phi_a(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r}, \tag{21}$$

$$E_\mathrm{H}[n] = \frac{1}{2}\iint n(\boldsymbol{r})w(\boldsymbol{r},\boldsymbol{r}')n(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}\,\mathrm{d}\boldsymbol{r}'. \tag{22}$$

Here T_s is the noninteracting kinetic energy and E_H is the classical electron–electron electrostatic energy. For the periodic MOF, w is the periodic Coulomb kernel. The exchange–correlation functional E_xc contains the remaining interaction energy and the correction to T_s. Thus the electronic energy functional becomes

$$E_\mathrm{el}[n;\boldsymbol{R}] = T_\mathrm{s}[n] + \int v_\mathrm{ext}(\boldsymbol{r};\boldsymbol{R})n(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} + E_\mathrm{H}[n] + E_\mathrm{xc}[n]. \tag{23}$$

Minimizing this energy with respect to the orbitals while preserving their orthonormality gives the one-electron KS equations:

$$\left[-\frac{\hbar^2}{2m_\mathrm{e}}\nabla^2 + v_\mathrm{eff}[n](\boldsymbol{r})\right]\phi_a(\boldsymbol{r}) = \epsilon_a\phi_a(\boldsymbol{r}), \tag{24}$$

$$v_\mathrm{eff}[n](\boldsymbol{r}) = v_\mathrm{ext}(\boldsymbol{r}) + v_\mathrm{H}[n](\boldsymbol{r}) + v_\mathrm{xc}[n](\boldsymbol{r}), \tag{25}$$

$$v_\mathrm{H}[n](\boldsymbol{r}) = \int w(\boldsymbol{r},\boldsymbol{r}')n(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}', \qquad v_\mathrm{xc}[n](\boldsymbol{r}) = \frac{\delta E_\mathrm{xc}}{\delta n(\boldsymbol{r})}. \tag{26}$$

The decomposition is formally exact when the auxiliary system can reproduce the density; practical DFT approximates E_xc. Because v_eff depends on n, the equations are solved self-consistently: start from a trial density, solve for the orbitals, rebuild the density, and iterate until both density and energy converge.

At the converged density n_0, adding nuclear repulsion gives the potential energy and its derivatives give the reference forces:

$$U_{\mathrm{DFT}}(\boldsymbol{R}) = E_{\mathrm{el}}[n_0; \boldsymbol{R}] + V_{\mathrm{nn}}(\boldsymbol{R}), \qquad \boldsymbol{F}_i^{\mathrm{DFT}} = -\nabla_{\boldsymbol{R}_i} U_{\mathrm{DFT}}|_{\boldsymbol{h}}\,. \tag{27}$$

Cell-strain derivatives similarly provide stresses. These quantities supply training labels for MLIPs such as PET-MAD [22]. In this project, the MLIP then evaluates the potential-energy surface during MD and Hessian calculations without repeating the electronic calculation.

2.3 Statistical Mechanics

The classical simulations in this project use the isothermal–isobaric (NPT) ensemble. The numbers of framework atoms and guest molecules are fixed, while the system exchanges energy and volume with reservoirs at temperature T and imposed hydrostatic pressure P. Allowing the periodic cell to fluctuate captures the volume response of the loaded framework. The quantity of interest is therefore the constant-pressure heat capacity at fixed loading [3, 4].

For N classical atoms, write the positions and momenta explicitly as $\boldsymbol{r}^N = (\boldsymbol{r}_1, \ldots, \boldsymbol{r}_N)$ and $\boldsymbol{p}^N = (\boldsymbol{p}_1, \ldots, \boldsymbol{p}_N)$. The cell dependence of functions is left implicit throughout this section: they are evaluated in the instantaneous periodic cell, with $V = \det \boldsymbol{h}$. The cell remains an independent fluctuating variable; positions and momenta alone do not determine its volume. We denote the physical Hamiltonian by E, reserving H for enthalpy:

$$E(\boldsymbol{r}^N, \boldsymbol{p}^N) = K(\boldsymbol{p}^N) + U(\boldsymbol{r}^N) = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m_i} + U(\boldsymbol{r}^N), \tag{28}$$

where U is the MLIP potential energy. The instantaneous physical enthalpy is

$$H(\boldsymbol{r}^N, \boldsymbol{p}^N) = E(\boldsymbol{r}^N, \boldsymbol{p}^N) + PV = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m_i} + U(\boldsymbol{r}^N) + PV. \tag{29}$$

It uses the imposed pressure and excludes thermostat and barostat energies.

With $\beta = (k_{\mathrm{B}}T)^{-1}$, the NPT partition function and probability density are

$$\Delta(N, P, T) = \mathcal{C} \int_0^\infty \frac{\mathrm{d}V}{V_0} \int_{\mathrm{cell}(V)} \mathrm{d}^{3N}r \int \mathrm{d}^{3N}p\, e^{-\beta H(\boldsymbol{r}^N, \boldsymbol{p}^N)}, \tag{30}$$

$$\rho(\boldsymbol{r}^N, \boldsymbol{p}^N) = \Delta^{-1} e^{-\beta H(\boldsymbol{r}^N, \boldsymbol{p}^N)}. \tag{31}$$

Here V_0 is a fixed, temperature- and pressure-independent reference volume, and $\mathcal{C}$ contains the fixed phase-space normalization factors needed to make Δ dimensionless. The position integral covers the instantaneous cell for every atom, and the momentum integral covers the allowed momenta. The volume integral is retained because the cell fluctuates in NPT. Although the cell argument is suppressed in ρ, this is the joint density of atomic positions, momenta, and cell volume, rather than a density with the volume already integrated out. Every equilibrium observable is averaged as

$$\langle A \rangle = \mathcal{C} \int_0^\infty \frac{\mathrm{d}V}{V_0} \int_{\mathrm{cell}(V)} \mathrm{d}^{3N}r \int \mathrm{d}^{3N}p\, \rho(\boldsymbol{r}^N, \boldsymbol{p}^N) A(\boldsymbol{r}^N, \boldsymbol{p}^N), \tag{32}$$

Throughout this report, P denotes the imposed hydrostatic pressure, and equilibrium averages, variances, and covariances refer to the NPT ensemble unless stated otherwise. Ensemble subscripts are omitted.

The corresponding thermodynamic potential is the Gibbs free energy. For a temperature-independent Hamiltonian,

$$G = -k_{\mathrm{B}}T \ln \Delta, \qquad \langle H \rangle = -\left(\frac{\partial \ln \Delta}{\partial \beta}\right)_{N,P}, \tag{33}$$

$$\langle V \rangle = \left(\frac{\partial G}{\partial P}\right)_{N,T}, \qquad S = -\left(\frac{\partial G}{\partial T}\right)_{N,P}, \tag{34}$$

$$\langle E \rangle = \langle H \rangle - P\langle V \rangle, \qquad G = \langle H \rangle - TS. \tag{35}$$

The volume integral above displays the scalar-volume formulation. In the actual simulations the cell matrix $\boldsymbol{h}$ also permits anisotropic strain under hydrostatic external pressure, with $V = \det \boldsymbol{h}$. Fractional coordinates $\boldsymbol{s}_i = \boldsymbol{h}^{-1}\boldsymbol{r}_i$ give $\mathrm{d}^{3N}r = V^N \mathrm{d}^{3N}s$; integrating out the momenta therefore gives a configurational weight proportional to $V^N e^{-\beta[U+PV]}$ in fractional coordinates. Flexible-cell sampling must include the appropriate cell measure and extended-system corrections [25].

Additionally, we assume the ergodic hypothesis, which states that the infinite time average of an observable along a trajectory is equal to its ensemble average. So the integral in Eq. (32) can be replaced by a time average along a sufficiently long trajectory:

$$\langle A \rangle = \lim_{\tau \to \infty} \frac{1}{\tau} \int_0^\tau A\big(\boldsymbol{r}^N(t), \boldsymbol{p}^N(t)\big)\, \mathrm{d}t. \tag{36}$$

In practice, we can't run the simulation for ever, so we need to run the simulation for long enough to ensure that the system has reached equilibrium and explored the relevant phase space. We can then replace the integral by a finite sum over the trajectory frames, giving the time average

$$\langle A \rangle \approx \frac{1}{M} \sum_{j=1}^{M} A\big(\boldsymbol{r}^N(t_j), \boldsymbol{p}^N(t_j)\big), \tag{37}$$

2.3.1 Equipartition and temperature

The classical equipartition theorem relates temperature to the mean energy stored in quadratic degrees of freedom: each independent term of the form $ax^2/2$, with $a > 0$, contributes $k_\mathrm{B}T/2$. The coefficient changes the amplitude of the fluctuations, but not their mean energy. This provides both a temperature estimator for MD and the classical reference for vibrational heat capacity [3, 4].

To derive the result, take an unconstrained momentum component or a harmonic displacement x ranging over the real line. Its conditional equilibrium density is proportional to $e^{-\beta a x^2/2}$: the remaining energy terms, including PV, are independent of x and cancel in the normalized average. Integration by parts gives

$$0 = \int_{-\infty}^{\infty} \frac{\mathrm{d}}{\mathrm{d}x}\left(x e^{-\beta a x^2/2}\right) \mathrm{d}x = \int_{-\infty}^{\infty} (1 - \beta a x^2) e^{-\beta a x^2/2}\, \mathrm{d}x, \tag{38}$$

$$\left\langle \frac{a x^2}{2} \right\rangle = \frac{1}{2\beta} = \frac{1}{2} k_\mathrm{B} T. \tag{39}$$

The boundary term vanishes because the Gaussian decays at infinity. The same result holds after averaging over the other variables, including the fluctuating cell. Its application to potential energy requires the quadratic approximation; its application to the separable kinetic energy does not require harmonic atomic interactions.

For an atomic momentum component, $x = p_{i\alpha}$ and $a = 1/m_i$. Summing over f active momentum degrees of freedom gives

$$\langle K \rangle = \frac{f}{2} k_\mathrm{B} T, \qquad T_\mathrm{kin} = \frac{2K}{f k_\mathrm{B}}. \tag{40}$$

Subtract constraints and deliberately removed momentum components from $3N$; fixing total linear momentum gives $f = 3N - 3$. Auxiliary reservoir variables do not count. Initial velocities follow the Maxwell–Boltzmann distribution:

$$v_{i\alpha} \sim \mathcal{N}\left(0, \frac{k_\mathrm{B} T}{m_i}\right), \qquad \alpha \in \{x, y, z\}. \tag{41}$$

Subtract center-of-mass velocity when required. The NPT momentum distribution gives $\mathrm{var}(K) = f(k_\mathrm{B}T)^2/2$ and $\mathrm{var}(T_\mathrm{kin}) = 2T^2/f$; repeated exact rescaling suppresses these fluctuations.

In this project, these relations set the initial velocity distribution and the kinetic-energy target of the thermostat, and provide a check on sampled temperature fluctuations. For each harmonic mode, equipartition assigns $k_\mathrm{B}T/2$ to kinetic energy and the same amount to potential energy, giving a classical heat capacity of k_B per mode. This is the harmonic contribution subtracted in the hybrid correction. Quantum modes with $\hbar\omega \geq k_\mathrm{B}T$, especially bond stretches, contribute less than this classical value. Section 2.3.3 derives their quantum contribution and the Dulong–Petit limit. The full classical C_P still comes from the sampled enthalpy: equipartition alone does not determine anharmonic or pressure–volume contributions.

2.3.2 Heat capacity from mean enthalpy

At fixed composition and imposed pressure, heat capacity measures the change in equilibrium mean enthalpy with temperature:

$$C_P(T) = \left(\frac{\partial \langle H \rangle}{\partial T}\right)_{N,P}, \qquad \langle H \rangle_T = \langle K + U + PV \rangle_T. \tag{42}$$

The project estimates this derivative from separate equilibrated MD trajectories at neighboring temperatures. On a uniformly spaced grid, the centered finite difference is

$$C_P(T) \simeq \frac{\langle H \rangle_{T+\Delta T} - \langle H \rangle_{T-\Delta T}}{2\Delta T}. \tag{43}$$

All trajectories entering the difference use the same composition, model, and imposed pressure. Each mean includes kinetic, potential, and pressure–volume contributions, retaining the anharmonic behavior sampled by classical MD. The mass-specific heat capacity is

$$c_P(T) = \frac{C_P(T)}{M_{\mathrm{cell}}}, \tag{44}$$

where M_{cell} includes both framework and guest mass.

For a smooth enthalpy curve, the centered approximation has a truncation error of order $(\Delta T)^2$. Smaller spacing reduces this error but amplifies sampling noise in the difference, so both trajectory sampling and temperature spacing require convergence checks. The production grid uses $\Delta T = 25\,\mathrm{K}$ and includes neighboring temperatures beyond the reported interval to retain centered derivatives at its endpoints (Section 4.3).

2.3.3 Quantum harmonic heat capacity and the Dulong–Petit law

Quantum statistical mechanics averages over populations of states as well as the measurement probabilities within each state. Here $\hat{E}$ is the Born–Oppenheimer nuclear Hamiltonian of Eq. (12); its position variables are the atomic coordinates denoted $\boldsymbol{r}^N$ in the classical discussion. At a fixed cell, let $\hat{E}|\chi_\nu\rangle = E_\nu|\chi_\nu\rangle$. If the system occupies energy eigenstate $|\chi_\nu\rangle$ with probability w_ν, then

$$\langle A\rangle_{\mathrm{q}} = \sum_\nu w_\nu \langle\chi_\nu|\hat{A}|\chi_\nu\rangle, \qquad \hat{\rho}_{\mathrm{q}} = \sum_\nu w_\nu |\chi_\nu\rangle\langle\chi_\nu|, \qquad \sum_\nu w_\nu = 1. \tag{45}$$

The density operator $\hat{\rho}_{\mathrm{q}}$ packages the statistical mixture [1]. For any complete orthonormal basis $\{|m\rangle\}$, define $\mathrm{Tr}\,\hat{B} = \sum_m \langle m|\hat{B}|m\rangle$. Inserting the density operator and using completeness gives

$$\mathrm{Tr}(\hat{\rho}_{\mathrm{q}}\hat{A}) = \sum_{m,\nu} w_\nu \langle m|\chi_\nu\rangle\langle\chi_\nu|\hat{A}|m\rangle \tag{46}$$

$$= \sum_\nu w_\nu \langle\chi_\nu|\hat{A}|\chi_\nu\rangle = \langle A\rangle_{\mathrm{q}}. \tag{47}$$

The trace is independent of basis, and $\hat{A}$ need not commute with $\hat{E}$. In the nuclear position basis the same average is

$$\langle A\rangle_{\mathrm{q}} = \int \mathrm{d}^{3N}r\, \langle \boldsymbol{r}^N|\hat{\rho}_{\mathrm{q}}\hat{A}|\boldsymbol{r}^N\rangle. \tag{48}$$

For a position-dependent observable, this reduces to the integral of $A(\boldsymbol{r}^N)$ weighted by $\langle \boldsymbol{r}^N|\hat{\rho}_{\mathrm{q}}|\boldsymbol{r}^N\rangle$.

At fixed cell and temperature, the Boltzmann populations give

$$w_\nu = \frac{e^{-\beta E_\nu}}{Z_{\mathrm{q}}}, \qquad Z_{\mathrm{q}} = \sum_\nu e^{-\beta E_\nu} = \mathrm{Tr}\, e^{-\beta\hat{E}}, \qquad \hat{\rho}_{\mathrm{q}} = \frac{e^{-\beta\hat{E}}}{Z_{\mathrm{q}}}. \tag{49}$$

The operator exponential is diagonal in the energy eigenbasis, which explains the equivalence between the sum and trace forms of the partition function. For a fluctuating cell, include the pressure work and integrate over volume as in the classical ensemble:

$$\Delta_{\mathrm{q}} = \int_0^\infty \frac{\mathrm{d}V}{V_0}\, \mathrm{Tr}\, e^{-\beta(\hat{E}+PV)}, \qquad \hat{\rho} = \Delta_{\mathrm{q}}^{-1} e^{-\beta(\hat{E}+PV)}, \tag{50}$$

$$\langle A\rangle = \int_0^\infty \frac{\mathrm{d}V}{V_0}\, \mathrm{Tr}(\hat{\rho}\hat{A}), \qquad 1 = \int_0^\infty \frac{\mathrm{d}V}{V_0}\, \mathrm{Tr}\,\hat{\rho}. \tag{51}$$

Here PV multiplies the identity operator and the cell dependence remains implicit. The trace is over nuclear states at each cell; the volume integral also averages over cells. For this project, the quantum correction is evaluated at a fixed optimized cell with fixed harmonic frequencies. The following mode averages therefore give C_V, while the classical MD contribution remains at constant pressure.

Expanding the potential about a stable minimum and using mass-weighted normal coordinates Q_k gives independent harmonic oscillators:

$$\hat{E}_{\mathrm{har}} = U_{\mathrm{min}} + \sum_{k\in\mathcal{V}} \left(\frac{\hat{\Pi}_k^2}{2} + \frac{\omega_k^2 \hat{Q}_k^2}{2} \right), \qquad \epsilon_{k,n} = \hbar\omega_k \left(n + \frac{1}{2} \right), \quad n = 0, 1, \ldots. \tag{52}$$

Here $\mathcal{V}$ contains the retained positive-frequency modes and Π_k is conjugate to Q_k. Each mode has partition function

$$z_k = \sum_{n=0}^\infty e^{-\beta\epsilon_{k,n}} = \frac{e^{-x_k/2}}{1-e^{-x_k}}, \qquad x_k = \beta\hbar\omega_k, \qquad Z_{\mathrm{har}} = e^{-\beta U_{\mathrm{min}}} \prod_{k\in\mathcal{V}} z_k. \tag{53}$$

Differentiating its logarithm gives the mean energy:

$$\langle E\rangle_{\mathrm{har}} = -\frac{\partial \ln Z_{\mathrm{har}}}{\partial \beta} = U_{\mathrm{min}} + \sum_{k\in\mathcal{V}} \hbar\omega_k \left(\frac{1}{2} + \frac{1}{e^{x_k}-1}\right). \tag{54}$$

The zero-point energy $\hbar\omega_k/2$ is independent of temperature at fixed frequencies and does not contribute to the heat capacity. Differentiating the thermal occupation gives

$$C_{V,\mathrm{q}}^{\mathrm{har}} = \left(\frac{\partial\langle E\rangle_{\mathrm{har}}}{\partial T}\right)_V = \sum_{k\in\mathcal{V}} k_{\mathrm{B}} \frac{x_k^2 e^{x_k}}{(e^{x_k}-1)^2}. \tag{55}$$

Modes with $\hbar\omega_k \gg k_{\mathrm{B}}T$ are thermally suppressed, whereas low-frequency modes approach their classical contribution.

In the high-temperature limit, $x_k \ll 1$, using $e^{x_k}-1 \simeq x_k$ gives $C_{V,k} \to k_{\mathrm{B}}$. Classical equipartition gives the same result: each oscillator has one quadratic kinetic and one quadratic potential term, together contributing $k_{\mathrm{B}}T$ to its mean energy. Thus

$$C_{V,\mathrm{cl}}^{\mathrm{har}} = |\mathcal{V}|k_{\mathrm{B}}, \qquad C_V \longrightarrow 3Nk_{\mathrm{B}}, \qquad C_{V,\mathrm{molar}} \longrightarrow 3R, \quad R = N_{\mathrm{A}}k_{\mathrm{B}}. \tag{56}$$

The last two expressions are the Dulong–Petit law for a bulk solid with N atoms, with molar heat capacity defined per mole of atoms. A finite periodic-cell calculation omitting the three uniform translations instead has $3N-3$ vibrational modes before any additional mode filtering. For the loaded MOF, N includes both framework and guest atoms; the mass-specific limit is $|\mathcal{V}|k_{\mathrm{B}}/M_{\mathrm{cell}}$.

The project uses these results to replace the classical harmonic contribution by its quantum value:

$$C_P^{\mathrm{hyb}}(T) \simeq C_{P,\mathrm{cl}}^{\mathrm{anh}}(T) + \sum_{k\in\mathcal{V}} k_{\mathrm{B}} \left[\frac{x_k^2 e^{x_k}}{(e^{x_k}-1)^2} - 1\right]. \tag{57}$$

The correction vanishes in the classical high-temperature limit. Adding this fixed-cell C_V correction to the MD C_P assumes that the quantum change in $C_P - C_V$ is small.

2.3.4 Classical MD and quantum nuclear statistics

On the Born–Oppenheimer surface introduced in Section 2.1.3, classical MD replaces nuclear operators by positions and momenta, giving the Hamiltonian in Eq. (28) and Newtonian forces $m_i\ddot{\boldsymbol{R}}_i = -\nabla_i U$. A useful harmonic criterion and thermal length scale are

$$\beta\hbar\omega \ll 1, \qquad \lambda_i = \sqrt{\frac{2\pi\hbar^2}{m_i k_{\mathrm{B}}T}}, \qquad \beta = (k_{\mathrm{B}}T)^{-1}. \tag{58}$$

Classical sampling is appropriate when quantum level spacing is small relative to thermal energy and nuclear delocalization is negligible on relevant structural scales. A hydrogen-containing system can satisfy these conditions for slow collective modes while violating them for bond vibrations.

The fixed-cell quantum averages are given by Eqs. (47) and (49). The harmonic derivation in Section 2.3.3 provides the quantum nuclear populations used in this project, while MD samples the classical anharmonic surface. Quantum anharmonic sampling is an extension beyond the reported calculations.

2.3.5 Trajectory analysis and error estimation

Only equilibrated trajectory segments are used for averages. Energy, volume, temperature, and structural observables should fluctuate without systematic drift before data collection begins. For M equally spaced samples of an observable A, the time average is

$$\overline{A} = \frac{1}{M}\sum_{n=1}^{M} A_n. \tag{59}$$

Successive samples are correlated. Their autocovariance and normalized autocorrelation at lag k are

$$C_A(k) = \langle (A_n - \langle A\rangle)(A_{n+k} - \langle A\rangle)\rangle, \tag{60}$$

$$\rho_A(k) = C_A(k)/C_A(0). \tag{61}$$

These correlations determine how much independent information a trajectory contains. The sample-mean variance is

$$\operatorname{var}(\overline{A}) = \frac{C_A(0)}{M}\left[1 + 2\sum_{k=1}^{M-1}\left(1 - \frac{k}{M}\right)\rho_A(k)\right]. \tag{62}$$

Estimate or truncate the correlation sum statistically. Effective sample counts depend on the observable; finer output stride does not add independent information over a fixed duration.

Block errors should plateau for blocks longer than the correlation time, while retaining enough blocks [15]. Independent replicas help detect slow modes, but neither method reveals a basin never visited. Resample whole replicas or sufficiently long blocks and recompute the full estimator, preserving cross-temperature correlations. Variance-based heat capacities require errors of squared fluctuations, not merely errors of the mean enthalpy. Sampling error, finite-size effects, integration bias, and potential error need separate checks.

2.4 Molecular dynamics

MD turns the potential-energy surface into a trajectory of atomic positions and momenta. The framework and guests evolve together, sampling vibrations, guest rearrangements, and cell deformation. Evaluating the full potential at every step includes classical anharmonic motion. Its equilibrium averages provide the enthalpies needed for C_P; the separate harmonic calculation supplies the quantum correction [3, 4].

2.4.1 Forces and time integration

At fixed cell, Newton's equations are

$$\dot{\boldsymbol{R}}_i = \frac{\boldsymbol{p}_i}{m_i}, \qquad \dot{\boldsymbol{p}}_i = \boldsymbol{F}_i, \qquad \boldsymbol{F}_i = -\nabla_{\boldsymbol{R}_i} U(\boldsymbol{R}; \boldsymbol{h}). \tag{63}$$

Here $\boldsymbol{R}$ collects all atomic positions, using the same coordinates written as $\boldsymbol{r}^N$ in Section 2.3. A force evaluation supplies the accelerations, which advance the system to its next configuration. Velocity Verlet implements this cycle with two momentum half steps:

$$\boldsymbol{p}_i^{n+1/2} = \boldsymbol{p}_i^n + \frac{\Delta t}{2}\boldsymbol{F}_i(\boldsymbol{R}^n), \tag{64}$$

$$\boldsymbol{R}_i^{n+1} = \boldsymbol{R}_i^n + \Delta t \frac{\boldsymbol{p}_i^{n+1/2}}{m_i}, \tag{65}$$

$$\boldsymbol{p}_i^{n+1} = \boldsymbol{p}_i^{n+1/2} + \frac{\Delta t}{2}\boldsymbol{F}_i(\boldsymbol{R}^{n+1}). \tag{66}$$

The method is second order, reversible, and symplectic for the fixed-cell Hamiltonian problem, requiring one new force evaluation per step. The production integrator adds thermostat and cell updates to this splitting. Its timestep must resolve the fastest flexible vibrations, particularly X–H stretches. For a harmonic mode, $\omega_{\max}\Delta t < 2$ is a stability condition, not an accuracy test: enthalpies and structural observables must also converge with timestep. Short fixed-cell energy-conservation tests help detect integration errors that a thermostat could otherwise mask.

2.4.2 Periodic geometry and interaction range

A periodic cell represents a repeating bulk material (Figure 2). With cell matrix $\boldsymbol{h}$, $V = \det \boldsymbol{h}$ and $\boldsymbol{R}_i = \boldsymbol{h}\boldsymbol{s}_i$, where $\boldsymbol{s}_i$ is fractional position. Periodic neighbors are separated by

$$\boldsymbol{r}_{ij\boldsymbol{n}} = \boldsymbol{R}_j - \boldsymbol{R}_i + \boldsymbol{h}\boldsymbol{n}, \qquad \boldsymbol{n} \in \mathbb{Z}^3. \tag{67}$$

Thus atoms near opposite faces can be close neighbors. In a one-dimensional cell of length 20 Å, positions 1 and 19 Å are only 2 Å apart through a periodic image. Crossing a boundary changes an atom's displayed image; unwrapped positions retain its displacement history.

Periodicity specifies geometry, not interaction range. The present PET model uses local, many-body neighborhoods; there is no separate long-range electrostatic solver in the MD input. For a short enough cutoff, the closest periodic image suffices. Otherwise multiple images within the cutoff must be included. A neighbor list adds a skin beyond the physical cutoff and is rebuilt as atoms or the cell move. Smooth cutoffs and correct triclinic image handling are essential for continuous forces.

2.4.3 Temperature and pressure control

Newtonian dynamics in a fixed periodic cell conserves the physical Hamiltonian $E = K + U$: it describes an isolated system at fixed volume. Assigning initial velocities sets an initial kinetic temperature, but does not maintain the desired temperature as potential and kinetic energy interconvert. For this project, we instead need equilibrium

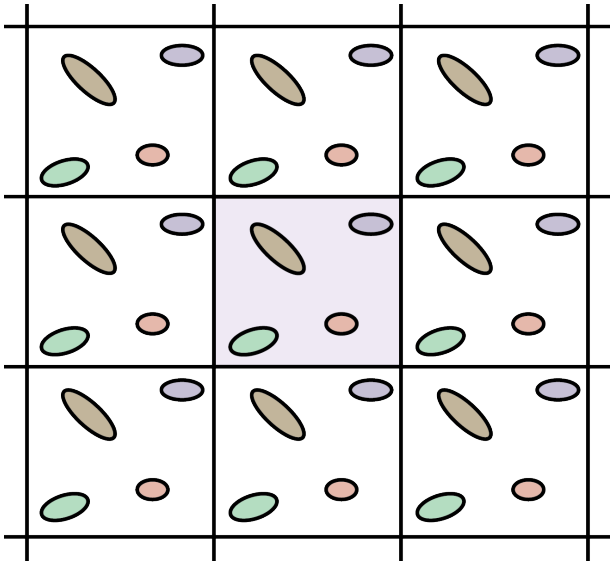


Figure 2: The central simulation cell and its periodic images in two dimensions. Neighbors can lie across a cell boundary. Reproduced unchanged from Christopher Rowley [10], CC BY-SA 4.0.

averages at prescribed T and P, including thermal expansion of the porous framework. A thermostat represents heat exchange with a reservoir; a barostat allows the cell to exchange mechanical work with its surroundings. Together they sample the distribution proportional to $\exp[-\beta(E + PV)]$ introduced in Section 2.3.

A thermostat couples the atomic momenta to auxiliary dynamical variables. A single Nosé–Hoover thermostat introduces an auxiliary friction variable:

$$\dot{\boldsymbol{p}}_i = \boldsymbol{F}_i - \xi \boldsymbol{p}_i, \qquad \dot{\xi} = \frac{2K - f k_{\mathrm{B}} T}{Q}, \qquad Q \sim f k_{\mathrm{B}} T \tau_T^2. \tag{68}$$

Here f is the number of active kinetic degrees of freedom, Q is the thermostat inertia, and τ_T is its characteristic response time. The sign of the feedback removes excess kinetic energy or restores it. For example, $2K > f k_{\mathrm{B}} T$ increases ξ, producing friction. Nosé–Hoover chains also thermostat the auxiliary variables, improving sampling of nearly harmonic systems [26]. The target temperature sets an equilibrium distribution, so instantaneous temperature should fluctuate rather than remain exactly fixed.

The physical potential U is unchanged, but the atomic system now exchanges energy with additional degrees of freedom. Introduce a dimensionless thermostat coordinate η and its momentum $p_\eta = Q\dot{\eta} = Q\xi$. The corresponding extended energy is

$$E_{\mathrm{ext},T}(\boldsymbol{R}, \boldsymbol{p}, \eta, p_\eta) = \underbrace{\sum_i \frac{|\boldsymbol{p}_i|^2}{2m_i} + U(\boldsymbol{R})}_{E} + \underbrace{\frac{p_\eta^2}{2Q}}_{\text{reservoir kinetic energy}} + \underbrace{f k_{\mathrm{B}} T \eta}_{\text{reservoir potential}}. \tag{69}$$

Energy lost by the atoms is gained by the reservoir, giving $\mathrm{d}E_{\mathrm{ext},T}/\mathrm{d}t = 0$ for the continuous equations. This quantity is often called an extended Hamiltonian. In the physical-time Nosé–Hoover formulation used here, it is a conserved energy rather than a canonical Hamiltonian generating all the equations through ordinary Hamilton equations [26, 12].

The barostat responds to instantaneous pressure, which contains kinetic and configurational contributions:

$$\boldsymbol{P}_{\mathrm{inst}} = \frac{1}{V}\left[\sum_i \frac{\boldsymbol{p}_i \otimes \boldsymbol{p}_i}{m_i} + \boldsymbol{W}\right], \qquad P_{\mathrm{inst}} = \frac{1}{3}\operatorname{tr} \boldsymbol{P}_{\mathrm{inst}}. \tag{70}$$

For the many-body model, the virial follows from the energy response to strain at fixed fractional positions:

$$W_{\alpha\beta} = -\left.\frac{\partial U}{\partial \epsilon_{\alpha\beta}}\right|_{\boldsymbol{\epsilon}=0, \{\boldsymbol{s}_i\}}, \qquad \boldsymbol{h} \mapsto (\boldsymbol{I} + \boldsymbol{\epsilon})\boldsymbol{h}. \tag{71}$$

The model's configurational stress excludes the kinetic term; tensile-positive stress has the opposite sign to pressure [11]. Hydrostatic loading imposes $P\boldsymbol{I}$ with zero shear target. A flexible barostat permits independent cell lengths and tilts, so an isotropic pressure need not produce isotropic deformation. Cell changes carry atoms through $\boldsymbol{R}_i = \boldsymbol{h}\boldsymbol{s}_i$, alongside their motion within the cell. The MTK corrections ensure the intended ensemble measure [25, 12].

The barostat adds the cell matrix $\boldsymbol{h}$ and six generalized cell momenta $\pi_a = W_a \nu_a$, where ν_a are the three normal and three shear strain-rate variables and W_a their fictitious inertias. Pressure imbalance drives these variables: excess internal pressure drives expansion, while pressure below the target drives contraction. Thermostat chains introduce coordinates and momenta (η_j, p_{η_j}) for the atoms and (ζ_j, p_{ζ_j}) for the barostat, with inertias Q_j and Q'_j. For the project's two chains of length three and six independently barostatted cell components, the extended energy becomes

$$\begin{aligned} E_{\mathrm{ext},TP} = \underbrace{\sum_i \frac{|\boldsymbol{p}_i|^2}{2m_i} + U(\boldsymbol{R};\boldsymbol{h})}_{E} + \underbrace{P \det \boldsymbol{h}}_{PV} + \underbrace{\sum_{a=1}^{6} \frac{\pi_a^2}{2W_a}}_{\text{cell kinetic energy}} \\ + \sum_{j=1}^{3} \left(\frac{p_{\eta_j}^2}{2Q_j} + \frac{p_{\zeta_j}^2}{2Q'_j} \right) \\ + k_{\mathrm{B}} T \left(f\eta_1 + \eta_2 + \eta_3 + 6\zeta_1 + \zeta_2 + \zeta_3 \right). \end{aligned} \tag{72}$$

Here $p_{\eta_j} = Q_j \dot{\eta}_j$ and $p_{\zeta_j} = Q'_j \dot{\zeta}_j$. The coefficients f and 6 count the atomic and cell degrees of freedom coupled to each first thermostat; each later thermostat controls one preceding chain variable. The PV term accounts for work against the imposed pressure. Under the constant hydrostatic target, this expression agrees with the LAMMPS reservoir-energy terms up to an additive constant [13]. It is conserved by the continuous extended equations, while numerical integration introduces a timestep-dependent error. The cell and reservoir momenta are extended dynamical variables; ordinary Hamilton equations applied to this expression alone do not reproduce the implemented MTK dynamics.

Production uses LAMMPS `fix npt`: Nosé–Hoover chains of length three thermostat both the atoms and the barostat. Its flexible-cell equations follow Shinoda's formulation combining the MTK hydrostatic equations with Parrinello–Rahman strain dynamics. The `tri` option allows all six cell degrees of freedom to vary independently, with equal normal-pressure targets and zero shear targets; `mtk yes` enables the ensemble-measure correction [12]. The configured response times are $\tau_T = 100\,\mathrm{fs}$ and $\tau_P = 1000\,\mathrm{fs}$, at a target pressure of 1 bar (Section 4.3).

Thermostat and barostat damping times set their response rates. Excessively fast coupling can excite oscillations, while slow coupling delays relaxation. The physical energy E fluctuates as the reservoirs exchange energy and volume with the system; auxiliary reservoir energies are excluded from the physical enthalpy $H = E + PV$.

2.4.4 From initialization to equilibrium averages

The workflow starts from a periodically valid guest-loaded structure. A fixed-cell minimization removes high-energy contacts; thermal velocities are then drawn from the Maxwell–Boltzmann distribution. Minimization finds a local minimum and does not establish equilibrium at finite T and P. An initial trajectory interval is therefore discarded while the cell, framework, and guest distributions relax.

Production frames must show stationary enthalpy and structural observables. A stable kinetic temperature alone does not demonstrate exploration of slow adsorption or framework modes. Independent replicas help expose incomplete sampling, and autocorrelation limits the information in successive frames (Section 2.3). Mean enthalpies at neighboring temperatures then provide the classical heat-capacity derivative. The numerical settings, continuation procedure, and output selection are given in Section 4.3.

2.5 Metal–organic frameworks

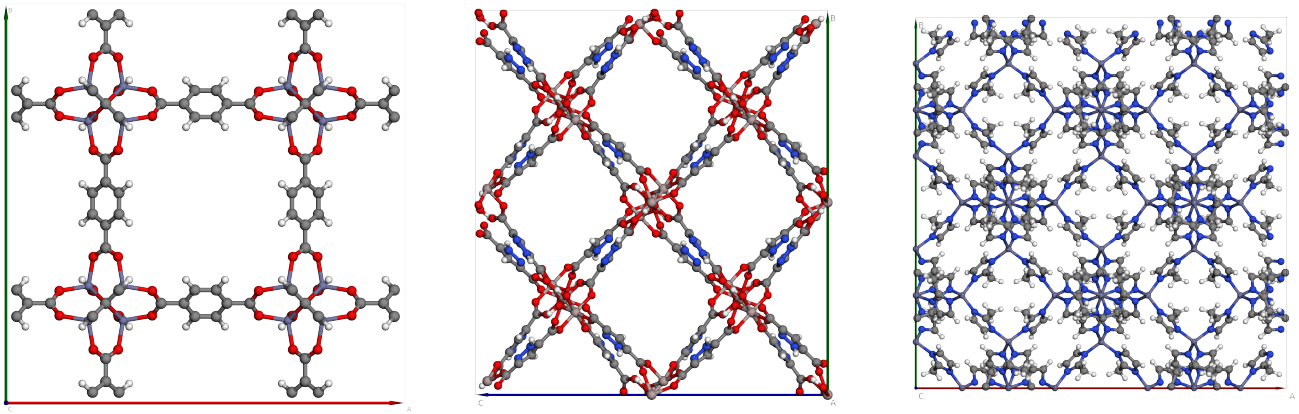
Metal–organic frameworks (MOFs) are porous crystals whose metal-containing nodes are connected by organic linkers. Their pore geometry and chemical functionalities determine where guest molecules bind and how they move. The simulations will compare MOF-5, MOF-303, and ZIF-8, illustrated using the empty project structures in Figure 3.

MOF-5 consists of Zn_4O nodes joined by benzene-1,4-dicarboxylate linkers, forming a three-dimensional network of pores. It provides the reference system for the methane-loading comparison with Kapil *et al.* [21].

MOF-303, Al(OH)(PZDC), contains aluminium–oxygen rods connected by 1H-pyrazole-3,5-dicarboxylate (PZDC) linkers. Its one-dimensional channels contain polar pyrazole and hydroxyl groups that provide hydrogen-bonding sites for water [8]. It therefore introduces a hydrophilic adsorption environment into the framework comparison.

ZIF-8, $Zn(mIm)_2$ with mIm denoting 2-methylimidazolate, consists of tetrahedrally coordinated zinc centres connected into a sodalite-type network. Its cages are connected by narrow windows; linker rotations can change their effective opening and couple guest adsorption to framework motion [9].

Across these frameworks, the heat capacity combines framework vibrations, guest internal modes, and redistribution among adsorption environments. Classical MD samples the anharmonic host–guest motion; normal modes of the optimized loaded structures provide the harmonic quantum correction. Differences in pore chemistry and flexibility motivate comparing all three hosts using the same thermodynamic workflow.



(a) MOF-5 (b) MOF-303 (c) ZIF-8

Figure 3: Empty framework starting structures rendered with Chemiscope from the project's structure dataset. MOF-5 contains 424 atoms; the MOF-303 and ZIF-8 $2\times2\times2$ supercells contain 1024 and 2208 atoms, respectively. Cell outlines indicate the periodic simulation boxes. Views are independently scaled to reveal the pore geometry.

2.6 Graph Theory

Graphs describe atomic neighborhoods and derivative dependencies [17, 20]. Their roles are distinguished in Table 1.

2.6.1 Adjacency, paths, and receptive fields

A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of vertices and edges. For an atomic graph, vertices represent atoms and edges identify neighbors within the model's interaction range. Periodic images carry translation labels, and messages may be directed even when the underlying neighbor relation is symmetric. For a simple undirected graph,

$$A_{ij} = \begin{cases} 1, & \{i,j\} \in \mathcal{E}, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathcal{N}_i = \{j : A_{ij} = 1\}. \tag{73}$$

Graph distance counts edges along a shortest path. The k-hop neighborhood

$$\mathcal{B}_k(i) = \{j : d_{\mathcal{G}}(i,j) \le k\} \tag{74}$$

contains all vertices reached in at most k steps. On the chain $1-2-3-4$, $\mathcal{B}_1(1) = \{1,2\}$ and $\mathcal{B}_2(1) = \{1,2,3\}$. This determines how far information propagates through a message-passing model.

Relabeling atoms permutes the adjacency matrix as $\boldsymbol{A} \mapsto \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{\mathrm{T}}$; the energy must remain unchanged, while atom-resolved outputs permute with the atoms. A separate computation graph records the operations used to evaluate the model. Its dependencies are differentiated by AD. Table 1 distinguishes these graphs from the derivative graphs used later in Section 2.8.2.

Table 1: Graph constructions used by the potential and its differentiation.

Graph	Vertices	Meaning of an edge
Interaction	Atoms	Neighbor relation, with periodic-image attributes
Computation	Values or operations	Directed computational dependence
Hessian sparsity	Coordinates (or atom blocks)	Possible off-diagonal curvature coupling
Column intersection	Derivative columns	Possible nonzeros sharing an output row

2.7 Machine-learned interatomic potentials

An MLIP learns a potential-energy surface from reference electronic-structure data, making long trajectories and Hessians affordable. Its usefulness requires both accurate predictions and coverage of the configurations visited by the loaded framework. This section introduces the PET model; checkpoint and conversion details belong to Section 4.2.

2.7.1 Energy, forces, and atomic graphs

An MLIP approximates the Born–Oppenheimer map $(\boldsymbol{R}, \boldsymbol{h}, \boldsymbol{Z}) \mapsto U$ with fitted parameters θ. A local, extensive decomposition is

$$U_\theta(\boldsymbol{R}, \boldsymbol{h}, \boldsymbol{Z}) = U_{\text{base}}(\boldsymbol{Z}) + \sum_{i=1}^{N} \varepsilon_i(\mathcal{N}_i; \theta), \tag{75}$$

The optional baseline depends only on composition. Atomic energies are model contributions, not unique observables. At bounded neighbor count, evaluation cost scales approximately linearly with atom count. Differentiating the same scalar energy gives conservative forces and a Cartesian Hessian:

$$\boldsymbol{F}_i = -\frac{\partial U_\theta}{\partial \boldsymbol{R}_i}, \tag{76}$$

Atoms are graph vertices with species features. Periodic directed edges $(v, u, \boldsymbol{n})$ carry relative geometry:

$$\boldsymbol{r}_{uv\boldsymbol{n}} = \boldsymbol{R}_v + \boldsymbol{h}\boldsymbol{n} - \boldsymbol{R}_u, \tag{77}$$

$$r_{uv\boldsymbol{n}} = \|\boldsymbol{r}_{uv\boldsymbol{n}}\| < r_{\text{c}}, \tag{78}$$

$$\mathcal{N}_u = \{(v, \boldsymbol{n}) : (v, u, \boldsymbol{n}) \in \mathcal{E}\}. \tag{79}$$

Retaining image labels distinguishes repeated neighbors. Species embeddings and radial edge features provide numerical inputs:

$$\boldsymbol{x}_u^{(0)} = \boldsymbol{E}_Z[Z_u] \in \mathbb{R}^d, \tag{80}$$

$$\boldsymbol{x}_{uv\boldsymbol{n}}^{\text{edge}} = f_{\text{c}}(r_{uv\boldsymbol{n}})\big[\varphi_1(r_{uv\boldsymbol{n}}), \ldots, \varphi_K(r_{uv\boldsymbol{n}})\big], \tag{81}$$

Directional models also use displacement vectors or their learned functions; latent feature vectors need not transform as physical vectors.

Figure 4 illustrates cutoff weighting. Smooth forces and Hessians require consistent derivatives through attention normalization and readout as well as radial features.

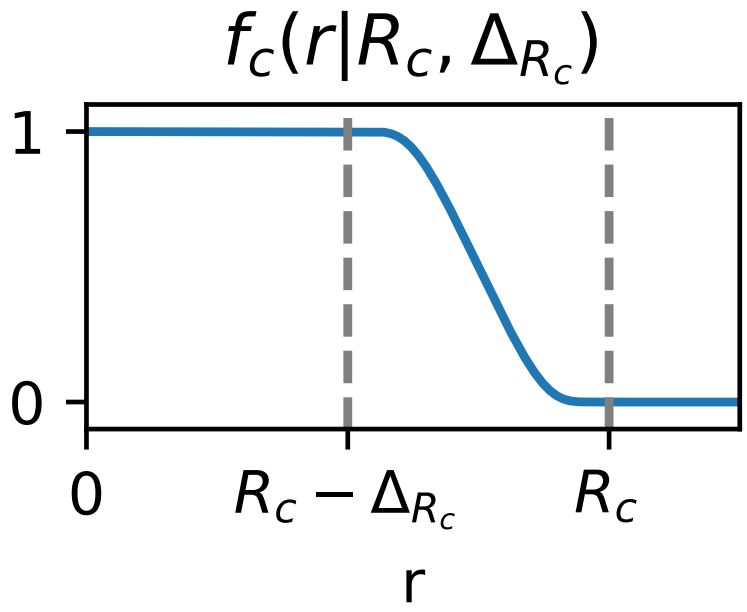


Figure 4: A smooth radial cutoff suppresses a neighbor contribution as its distance approaches the outer cutoff r_{c}. Smooth decay is needed for continuous energies, forces, and Hessians. Figure from the PET paper [23].

Node-wise F and scalar readout f obey

$$F(\boldsymbol{P}\boldsymbol{X}, \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{\mathrm{T}}) = \boldsymbol{P}F(\boldsymbol{X}, \boldsymbol{A}), \tag{82}$$

$$f(\boldsymbol{P}\boldsymbol{X}, \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{\mathrm{T}}) = f(\boldsymbol{X}, \boldsymbol{A}). \tag{83}$$

Relabel edge attributes and periodic images consistently.

2.7.2 Message passing

Each layer combines incoming messages with a symmetric aggregation $\bigoplus$ (usually a sum), then updates the node representation [20]:

$$\boldsymbol{h}_u = \phi\left(\boldsymbol{x}_u, \bigoplus_{(v,\boldsymbol{n})\in\mathcal{N}_u} \psi\big(\boldsymbol{x}_u, \boldsymbol{x}_v, \boldsymbol{x}^{\text{edge}}_{uv\boldsymbol{n}}\big)\right). \tag{84}$$

Repeating the update through L layers creates many-body features over an L-hop receptive field. A shared readout $\varepsilon_u = \rho(\boldsymbol{h}_u^{(L)})$ gives the total energy through Eq. (75).

2.7.3 Point Edge Transformer architecture

PET retains distinct directed-edge messages while attention mixes all tokens in one atomic environment (Figure 5) [23]. It underlies PET-MAD [22]. In the original formulation,

$$\boldsymbol{p}_{ij}^{(\ell)} = \mathrm{SiLU}\big(\boldsymbol{W}_r^{(\ell)}\boldsymbol{r}_{ij} + \boldsymbol{b}_r^{(\ell)}\big), \tag{85}$$

$$\boldsymbol{t}_{ij}^{(\ell)} = \mathrm{MLP}_\ell\Big(\boldsymbol{m}_{j\to i}^{(\ell-1)} \| \boldsymbol{E}_n^{(\ell)}[Z_j] \| \boldsymbol{p}_{ij}^{(\ell)}\Big), \tag{86}$$

Here $\|$ concatenates the incoming message, neighbor embedding, and position encoding. The MLP maps $3d_{\mathrm{PET}}$ features to d_{PET}; the first block omits the incoming message. A separate central token is $\boldsymbol{t}_{i0}^{(\ell)} = \boldsymbol{E}_c^{(\ell)}[Z_i]$.

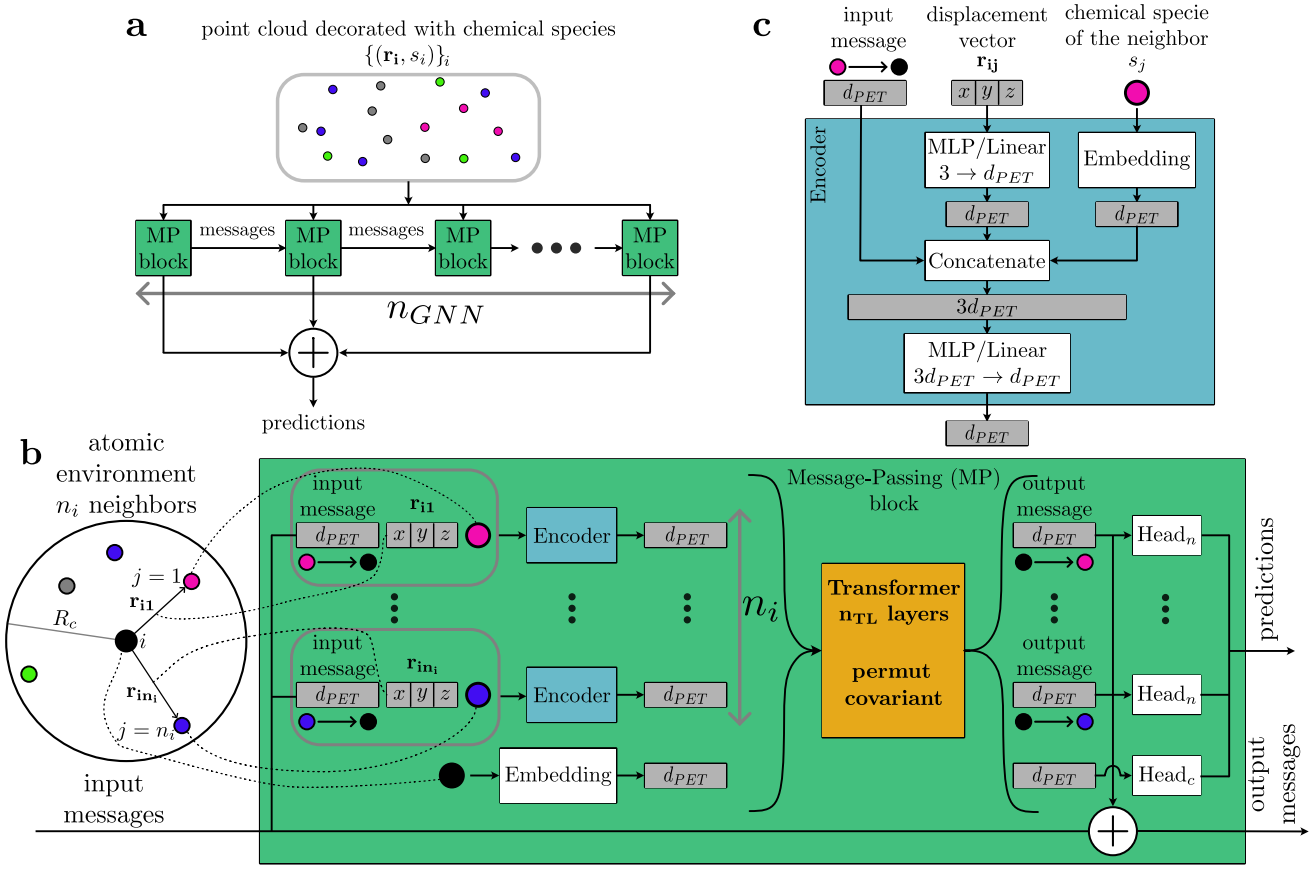


Figure 5: Point Edge Transformer (PET) message-passing architecture. Edge messages associated with a central atom are jointly processed by attention while retaining their individual edge identities for the outgoing update. Figure from the PET paper [23].

For one attention head,

$$\boldsymbol{q}_a = \boldsymbol{W}_Q \boldsymbol{t}_a, \qquad \boldsymbol{k}_b = \boldsymbol{W}_K \boldsymbol{t}_b, \tag{87}$$

$$\boldsymbol{v}_b = \boldsymbol{W}_V \boldsymbol{t}_b, \tag{88}$$

$$\alpha_{ab} = \frac{\exp(\boldsymbol{q}_a^{\mathrm{T}} \boldsymbol{k}_b / \sqrt{d_k}) c_b}{\sum_c \exp(\boldsymbol{q}_a^{\mathrm{T}} \boldsymbol{k}_c / \sqrt{d_k}) c_c}, \tag{89}$$

$$\boldsymbol{y}_a = \sum_b \alpha_{ab} \boldsymbol{v}_b. \tag{90}$$

With $c_0 = 1$ and $c_j = f_{\mathrm{c}}(r_{ij})$, departing neighbors vanish from both numerator and normalization. Multi-head attention, feed-forward layers, and residual connections preserve neighbor permutation equivariance. Attention layers enrich interactions within one environment; message-passing blocks extend the receptive field. Local attention cost is quadratic in neighbor count, but total cost remains approximately linear in atom count at fixed architecture and bounded neighborhoods.

Outgoing edge tokens update the messages. The original summed-edge readout is

$$\begin{aligned} U_\theta = \sum_i b_{Z_i} + \sum_{\ell=1}^{L} \sum_i \Big[& H_{\mathrm{c}}^{(\ell)}(\boldsymbol{x}_{i0}^{(\ell)}) \\ & + \sum_{j \in \mathcal{N}_i} f_{\mathrm{c}}(r_{ij}) H_n^{(\ell)}(\boldsymbol{x}_{ij}^{(\ell)}) \Big], \end{aligned} \tag{91}$$

H_c and H_n are learned heads, $\boldsymbol{x}$ are output tokens, and b_Z are species baselines. Every block contributes; cutoff weighting also enters the readout. Edge contributions are learned many-body terms, not pair energies. Project-specific dimensions appear in Section 4.2.

2.7.4 Physical symmetries

Scalar energies should respect translation, rotation/reflection, and identical-atom permutation:

$$\begin{aligned} U(\boldsymbol{R}) &= U(\boldsymbol{R} + \boldsymbol{a}) = U(\boldsymbol{Q}\boldsymbol{R}) \\ &= U(\boldsymbol{P}\boldsymbol{R}), \qquad \boldsymbol{Q} \in O(3). \end{aligned} \tag{92}$$

Forces then transform equivariantly, $\boldsymbol{F}_i(\boldsymbol{Q}\boldsymbol{R}) = \boldsymbol{Q}\boldsymbol{F}_i(\boldsymbol{R})$; periodic tests rotate both positions and cell. PET enforces translation and permutation symmetries, but learns rotational behavior through augmentation. Reported rotational errors below prediction errors [22] do not validate this checkpoint on loaded MOF-5.

2.7.5 Training and validity

A generic energy–force objective is

$$\begin{aligned} \mathcal{L}(\theta) = \sum_s \Bigg[& w_E \left(\frac{U_\theta^{(s)} - U_{\mathrm{ref}}^{(s)}}{N_s} \right)^2 \\ & + \frac{w_F}{3N_s} \sum_{i\alpha} \left(F_{i\alpha,\theta}^{(s)} - F_{i\alpha,\mathrm{ref}}^{(s)} \right)^2 \Bigg], \end{aligned} \tag{93}$$

The weights balance energy and force residuals; stress terms may also be included. This project uses pretrained PET models rather than fitting new potentials. PET-MAD was trained on a chemically diverse DFT dataset [22].

An MLIP inherits its reference method's approximations and may extrapolate poorly to unfamiliar environments. Hold-out errors, trajectory stability, and uncertainty indicators test different aspects of validity. Curvatures need explicit validation because the usual loss does not constrain Hessians directly; PET's methane benchmarks do not establish accuracy for adsorption in MOF-5.

2.8 Harmonic force constants and automatic differentiation

Let $\boldsymbol{R}_0$ be a stationary reference structure and $\boldsymbol{u} = \boldsymbol{R} - \boldsymbol{R}_0$. Expanding the potential energy gives

$$\begin{aligned} U(\boldsymbol{R}_0 + \boldsymbol{u}) = U(\boldsymbol{R}_0) + & \underbrace{\nabla U(\boldsymbol{R}_0) \cdot \boldsymbol{u}}_{=0 \text{ at a minimum}} \\ & + \frac{1}{2} \boldsymbol{u}^{\mathrm{T}} \boldsymbol{H} \boldsymbol{u} + \mathcal{O}(u^3), \end{aligned} \tag{94}$$

with Cartesian Hessian

$$H_{i\alpha,j\beta} = \frac{\partial^2 U}{\partial R_{i\alpha}\partial R_{j\beta}} = -\frac{\partial F_{i\alpha}}{\partial R_{j\beta}}. \tag{95}$$

For N atoms, $\boldsymbol{H}$ has dimensions $3N \times 3N$. Mass weighting connects its curvature to the normal-mode frequencies used in Section 2.3.3:

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}}, \qquad \boldsymbol{D}\boldsymbol{e}_k = \omega_k^2 \boldsymbol{e}_k. \tag{96}$$

Positive eigenvalues describe stable vibrations; negative eigenvalues indicate unstable directions and give imaginary frequencies. The Hessian symbol $\boldsymbol{H}$ denotes a matrix, distinct from the scalar enthalpy H.

2.8.1 Automatic differentiation

At fixed parameters, a neural-network energy composes differentiable operations:

$$\boldsymbol{z}^{(0)} = \boldsymbol{R}, \qquad \boldsymbol{z}^{(\ell+1)} = \phi_\ell(\boldsymbol{z}^{(\ell)}; \theta), \qquad U_\theta = \rho(\boldsymbol{z}^{(L)}). \tag{97}$$

AD applies their chain rules without a finite-displacement step [30]. It differentiates the implemented potential, up to floating-point error; it does not improve the potential's accuracy. For $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$,

$$(\boldsymbol{J}_f)_{pq} = \frac{\partial f_p}{\partial x_q}, \qquad \boldsymbol{J}_f \in \mathbb{R}^{m\times n}. \tag{98}$$

Forward AD evaluates $\boldsymbol{J}_f \boldsymbol{v}$ (JVP); reverse AD evaluates $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{J}_f$ (VJP). Dense recovery requires n forward or m reverse seeds, without storing layer Jacobians [31]. A single reverse sweep yields ∇U and hence all forces. Forward-over-reverse evaluates

$$\boldsymbol{H}\boldsymbol{v} = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\nabla_{\boldsymbol{R}} U_\theta(\boldsymbol{R} + \epsilon \boldsymbol{v})\Big|_{\epsilon=0}. \tag{99}$$

Dense Hessian recovery needs $3N$ such products. Sparse reconstruction reduces the seed count; cutoff smoothness and neighbor-selection conventions still control which function is differentiated.

2.8.2 Graph coloring and sparse automatic differentiation

A proper vertex coloring assigns an integer $c(q) \in \{1, \ldots, n_c\}$ to each vertex so that adjacent vertices have different colors. For differentiation, a color identifies input directions that can be processed together. The relevant edges must therefore represent conflicts in derivative recovery, rather than simply atomic neighbor relations.

ASD separates structural preparation from numerical differentiation [31]:

$$\underbrace{\text{detect pattern} \longrightarrow \text{color}}_{\text{preparation}} \longrightarrow \underbrace{\text{evaluate colored products} \longrightarrow \text{decompress}}_{\text{numerical evaluation}}. \tag{100}$$

Preparation can be reused while the pattern remains valid.

From derivatives to a conflict graph. For a function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, let $J_{pq} = \partial f_p/\partial x_q$. A structural sparsity pattern $\boldsymbol{S} \in \{0,1\}^{m\times n}$ satisfies

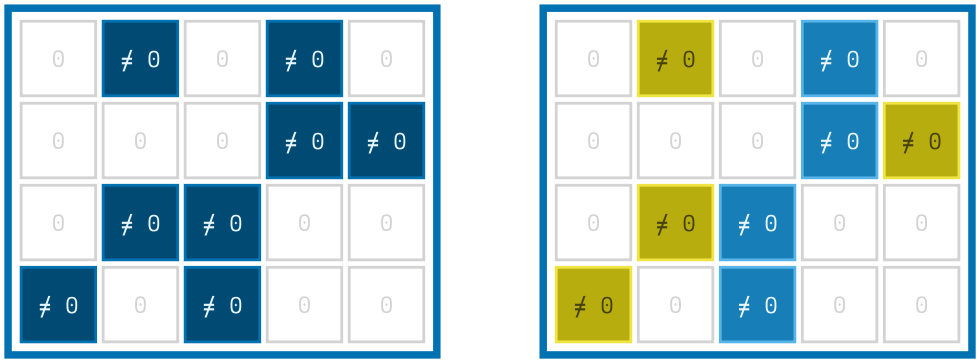
$$S_{pq} = 0 \implies J_{pq} = 0, \qquad \mathcal{R}_q = \{p : S_{pq} = 1\}. \tag{101}$$

Thus $\mathcal{R}_q$ lists the output rows that column q may affect. A numerical zero at one configuration does not necessarily imply a structural zero. Two columns conflict when their supports overlap, so the *column-intersection graph* has one vertex per input coordinate and edges

$$\{q, r\} \in \mathcal{E}_{\mathrm{col}} \iff q \neq r \text{ and } \mathcal{R}_q \cap \mathcal{R}_r \neq \varnothing. \tag{102}$$

Proper coloring of this graph ensures that columns of the same color have disjoint supports, also called structural orthogonality [31].

The pattern can be inferred by propagating input-dependency sets through the computation graph, without evaluating every derivative. For addition, for example, the dependency set of $z = u + v$ is the union of those of u and v. A global pattern covers the input domain; a local pattern may exploit the active branches at one input but requires revalidation [31].



(a) Sparsity pattern (b) Column groups

Figure 6: Structural preparation: permitted entries determine two groups of nonoverlapping columns. Reproduced from Hill, Dalle, and Montoison, Figure 12 [31].

Colored seeds, compression, and recovery. Forward AD evaluates a Jacobian-vector product without first forming $\boldsymbol{J}$. The ordinary seed $\boldsymbol{e}_q$ yields column q; using all n coordinate seeds would therefore require n directional evaluations. For a composition $\boldsymbol{f} = \boldsymbol{h} \circ \boldsymbol{g}$, the chain rule acts on a seed as

$$\boldsymbol{J}_f(\boldsymbol{x})\boldsymbol{v} = \boldsymbol{J}_h(\boldsymbol{g}(\boldsymbol{x}))\big[\boldsymbol{J}_g(\boldsymbol{x})\boldsymbol{v}\big]. \tag{103}$$

AD propagates the direction through these operations without storing the intermediate Jacobian matrices [30]. Construct one seed per color:

$$\boldsymbol{v}_a = \sum_{q:c(q)=a} \boldsymbol{e}_q, \qquad V_{qa} = \begin{cases} 1, & c(q) = a, \\ 0, & \text{otherwise}, \end{cases} \qquad \boldsymbol{Y} = \boldsymbol{J}\boldsymbol{V}. \tag{104}$$

Each compressed entry is a sum, $Y_{pa} = \sum_{q:c(q)=a} J_{pq}$. Structural orthogonality guarantees that at most one term is permitted to be nonzero in a given row. Consequently,

$$J_{pq} = Y_{p,c(q)} \quad \text{for } S_{pq} = 1. \tag{105}$$

The pattern identifies which column supplied each value. Sparse automatic differentiation (ASD) combines this graph-based preparation with ordinary AD and subsequent decompression; the AD still differentiates the original function. The number of directions falls from n to n_c, rather than computing a dense derivative and discarding its zeros afterward.

Figure 7 illustrates both numerical steps.

For the MOF Hessian, take $\boldsymbol{f} = \nabla U$ and $n = m = 3N$. Then $\boldsymbol{J}_f = \boldsymbol{H}$, and a forward derivative of the reverse-mode energy gradient gives $\boldsymbol{H}\boldsymbol{v}_a$ (Eq. (99)). Coloring therefore groups Cartesian coordinates whose Hessian columns can be recovered from the same Hessian-vector product.

Figure 8 connects graph edges to conflicts between derivative columns.

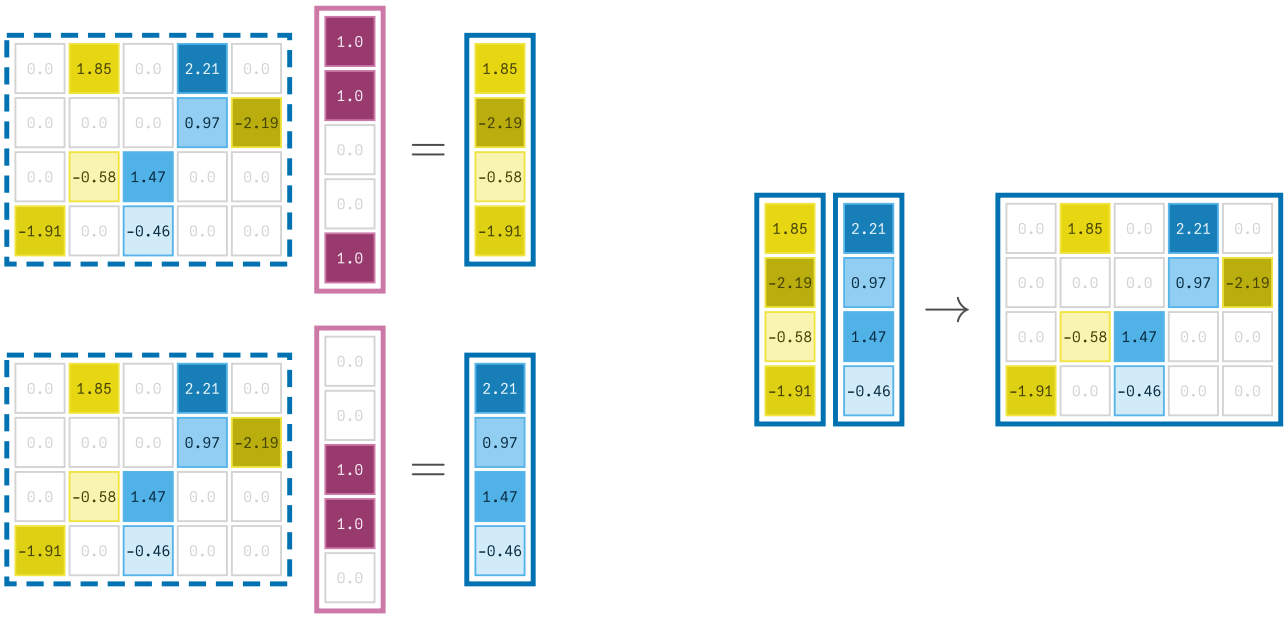
A worked example. Consider the symmetric tridiagonal Hessian

$$\boldsymbol{M} = \begin{pmatrix} a & b & 0 & 0 \\ b & c & d & 0 \\ 0 & d & e & f \\ 0 & 0 & f & g \end{pmatrix}. \tag{106}$$

Its off-diagonal sparsity graph is the chain $1 - 2 - 3 - 4$. Coloring that chain with $(1, 2, 1, 2)$ is proper, but does not permit direct column recovery: columns 1 and 3 both affect row 2, yielding the inseparable sum $b + d$. The column-intersection graph also connects vertices at distance two along the chain. A valid coloring of this larger conflict graph is $(1, 2, 3, 1)$, giving

$$\boldsymbol{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \boldsymbol{M}\boldsymbol{V} = \begin{pmatrix} a & b & 0 \\ b & c & d \\ f & d & e \\ g & 0 & f \end{pmatrix}. \tag{107}$$

The first product combines columns 1 and 4, whose supports are disjoint: rows 1–2 supply column 1 and rows 3–4 supply column 4. Three products recover every entry, whereas dense recovery would use four. For a longer tridiagonal matrix, repeating three colors still suffices, so the benefit grows with matrix size.



(a) Two colored JVPs

(b) Decompression

Figure 7: Five columns recovered from two directional evaluations using their known supports. Reproduced from Hill, Dalle, and Montoison, Figure 10 [31].

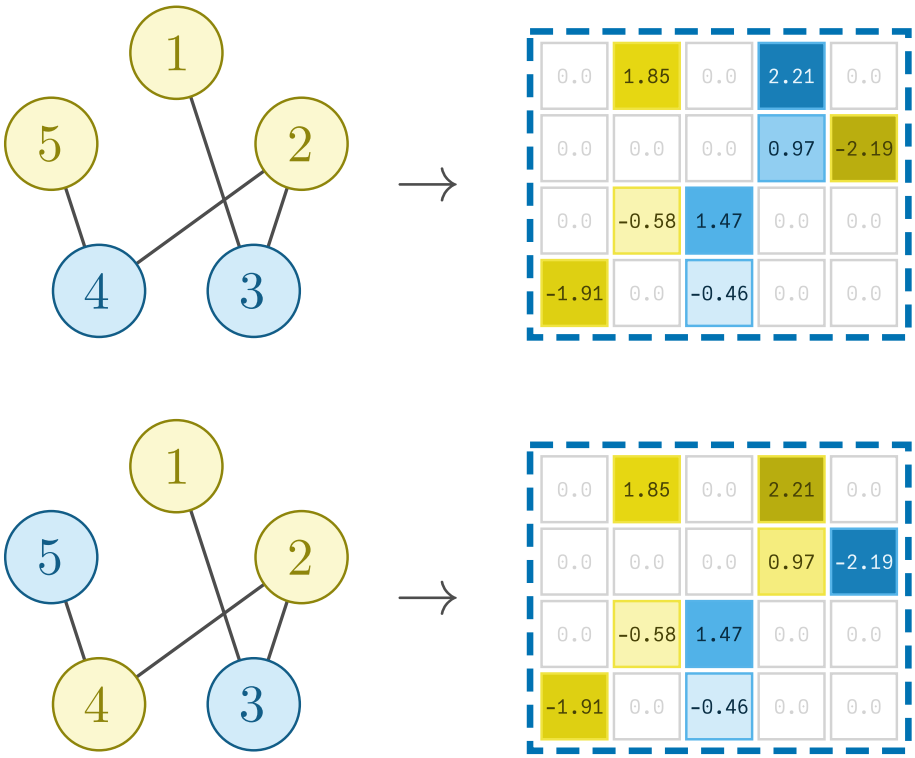


Figure 8: Valid coloring (top) and conflicting same-color columns (bottom). Reproduced from Hill, Dalle, and Montoison, Figures 17 and 19 [31].

Star coloring exploits Hessian symmetry. For a symmetric Hessian, $H_{pq} = H_{qp}$ supplies two possible compressed locations for each off-diagonal value: $Y_{p,c(q)}$ and $Y_{q,c(p)}$. It is sufficient that one of these is uncontaminated. This permits a less restrictive coloring than requiring every pair of column supports to be disjoint [18].

A *star coloring* is a proper coloring of the off-diagonal Hessian sparsity graph in which every path on four distinct vertices uses at least three colors. Equivalently, the subgraph formed by any two colors is a collection of stars. In each star, the center is the hub and its neighbors are spokes. A spoke has only one neighbor with the hub's color, so the hub–spoke entry can be extracted from that spoke's row; symmetry supplies the transposed entry. Proper coloring also keeps diagonal entries separate from off-diagonal contributions [19].

For the same chain, $(1, 2, 1, 3)$ is a star coloring. Its compression is

$$\boldsymbol{W} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{B} = \boldsymbol{M}\boldsymbol{W} = \begin{pmatrix} a & b & 0 \\ b+d & c & 0 \\ e & d & f \\ f & 0 & g \end{pmatrix}. \tag{108}$$

Although $B_{21} = b + d$ cannot be separated directly, $b = B_{12}$ and $d = B_{32}$ are individually available from the opposite orientations. The remaining values are $a = B_{11}$, $c = B_{22}$, $e = B_{31}$, $f = B_{33} = B_{41}$, and $g = B_{43}$. This example uses three colors with either strategy; star coloring can reduce the count further for other symmetric sparsity patterns.

Accuracy and cost. The coloring determines how many derivative directions are evaluated; Section 4.4 explains its application to PET's atom-pair sparsity pattern.

The pattern must include every actual coupling for exact recovery. For example, if the true matrix above also had $M_{24} = M_{42} = \delta$ but the pattern omitted this pair, the direct coloring would still combine columns 1 and 4 and incorrectly recover M_{21} as $b + \delta$. Thus an underestimated pattern can corrupt retained entries as well as omit missing ones. A greedy algorithm processes vertices in an ordering and assigns each the smallest admissible color; ordering affects the resulting count. For star coloring, admissibility also enforces the four-vertex-path rule. Greedy coloring need not find the minimum possible number of colors to give correct recovery; extra colors cost work [19].

3 Reference evidence and the hybrid approximation

Kapil *et al.* tested empty MOF-5 and 50-, 100-, and 150-methane cells [21]. Their hybrid strategy motivates this project; its accuracy on PET remains to be established.

3.1 Differences from the reference calculation

The reference used QuickFF [24]: B3LYP cluster-based bonded terms, MBIS charges from periodic PBE, Gaussian electrostatics, and MM3 Buckingham interactions with a 15 Å cutoff and tail corrections. RASPA insertion and canonical Monte Carlo preceded MD. Here PET supplies all energies, forces, and Hessians after independent random guest placement.

Both workflows differentiate neighboring equilibrium enthalpies using Eq. (43), with $\Delta T = 25\,\mathrm{K}$. The reference used optimized Yaff Hessians and TAMkin modes, compared QuickFF and UFF4MOF for the empty host, and optimized five Monte Carlo configurations for the 100-methane harmonic calculation.

3.2 Physical conclusions

The reference found nearly harmonic, quantum-mechanical empty MOF-5 (roughly 0.7 J/g/K near room temperature). Classical occupations overestimate its heat capacity. Methane introduces anharmonic host–guest motion and a loading-dependent minimum near 200 K, associated with changing attractive interactions as guests become mobile. The combination

$$C \approx C_{\mathrm{qn}}^{\mathrm{har}} - C_{\mathrm{cl}}^{\mathrm{har}} + C_{\mathrm{cl}}^{\mathrm{anh}} \tag{109}$$

reproduced the minimum qualitatively and was useful above about 200 K in their 100-methane test. A loaded-system spectrum is essential: the empty Hessian omits guest modes and host–guest coupling. Equation (127) gives the operational PET estimator.

4 Present computational methodology

4.1 Structure preparation and methane insertion

The conventional host has 424 atoms and cell parameter 25.866 Å. The campaign module `mof_heat_capacity.protocols.loaded` inserts centered methane molecules with uniform quaternion rotations and fractional positions,

rejecting periodic atom–atom overlaps. Independent insertion and velocity seeds define replicas. Random placement prevents severe contacts but is not an equilibrium adsorption distribution; subsequent equilibration must be validated.

4.2 PET model and implementation

The supplied PET-MAD and PET-SOL conversions share the architecture in Table 2; identical settings do not imply identical weights or training data. The original PET default used three attention layers per block, rather than one here.

Table 2: Architecture settings recorded in the local PET-MAD and PET-SOL PET-JAX metadata. The species count is embedding capacity, not a measure of training coverage.

Setting	Value
Outer cutoff	8.0 Å
Cutoff switching width	0.5 Å
Adaptive neighbor-selection setting	40
Message-passing blocks	3
Attention layers per block	1
Token width, d_{PET}	256
Attention heads	8
Transformer feed-forward width	512
Node feature width, `d_node`	1024
Readout feature width, `d_head`	256
Species-embedding capacity	102

The installed PyTorch implementation (`metatrain` 2026.3.1) embeds $(x_{ij}, y_{ij}, z_{ij}, r_{ij})$ and initializes edge messages with species embeddings. Central features contract to token width for attention and expand afterward. Attention-head width is $256/8 = 32$; `d_head=256` denotes readout width. Neighbor masking and log-cutoff logits implement Eq. (90). Reverse-neighbor routing and averaging with old messages update edge states.

The repository exports checkpoints with `upet.save_upet` for LAMMPS/metatomic MD, and converts matching weights to PET-JAX for SADMOF Hessians. Setup pins PET-JAX to commit `ed0add4`; conversion does not retrain. Composition baselines have zero Cartesian derivatives; energy scales must carry through forces and Hessians.

Compare energies and forces across export and conversion, including adaptive-cutoff derivative conventions. The sparse path stops those derivatives (Section 4.4). NPT additionally requires validated stress output on representative loaded-MOF structures; matching filenames or capability flags alone are insufficient [22].

4.3 Classical NPT molecular dynamics

Fresh loaded-system runs first use fixed-cell FIRE relaxation, with the force tolerance and optimization budgets recorded in the generated input. The maximum Cartesian force component must meet the tolerance before thermal velocities are assigned; failed convergence stops the run. The configured MD timestep is restored after FIRE's adaptive optimization. This initialization precedes equilibration and is distinct from the later Hessian quench [14].

The reported methane campaigns propagate loaded MOF-5. LAMMPS/metatomic uses a flexible triclinic cell with isotropic 1 bar target pressure and a Nosé–Hoover-chain/MTTK NPT fix. The campaign settings are

Table 3: Configured NPT campaign. Equilibration and convergence are assessed from observables, not assumed from these durations.

Parameter	Value
Particle / barostat thermostat chain lengths	3 / 3
Thermostat / barostat damping	100 / 1000 fs
Timestep / total steps	0.5 fs / 10^6
Trajectory / discarded interval	500 / 100 ps
Save interval	100 steps (0.05 ps)
Models / methane loadings	PET-MAD, PET-SOL / 50, 100, 150
MD grid	175–425 K, every 25 K
Hybrid reporting grid	200–400 K, every 25 K
Available replicas	One per state point

Production means use physical enthalpy,

$$\langle H \rangle_T = \langle E + PV \rangle_T \,, \tag{110}$$

then replica averaging, where available, and centered differentiation with neighbors separated by 50 K. This term retains sampled anharmonic motion. Independent replicas, longer production, and temperature-spacing convergence remain outstanding (Section 5.2).

Thermodynamic output includes potential and kinetic energies, cell volume, temperature, and the pressure tensor. Production trajectory output starts after the prescribed equilibration interval; logs and frames are matched by timestep. The input uses a 2 Å neighbor-list skin with rebuild checks every step and saves numeric restarts every 10^5 steps. Continuations restore the latest restart, skip minimization and velocity assignment, and advance to the original final-step target. They extend one trajectory rather than create independent replicas.

4.4 Sparse Hessian evaluation with SADMOF

SADMOF constructs an atom-pair pattern from PET's selected-neighbor graph, then expands each pair into a Cartesian 3×3 block [32]. For L message-passing layers, the maximum Hessian reach is $K = 2L$ for node readout and $K = 2L + 1$ for PET edge readout. Boolean adjacency gives

$$\boldsymbol{P}^{(k)} = \boldsymbol{I} \vee \boldsymbol{A} \vee \boldsymbol{A}^2 \vee \cdots \vee \boldsymbol{A}^k, \tag{111}$$

At $k = K$ the pattern contains all couplings under the chosen derivative convention; $k < K$ truncates it. Additional nonlocal terms need separate analysis. Structural sparsity requires

$$S_{pq} = 0 \quad \Longrightarrow \quad H_{pq} = 0 \tag{112}$$

An overestimate costs entries and colors. An underestimate can both discard curvature and contaminate retained entries. Mixed second derivatives depend on nonlinear readout, not just direct interaction edges.

ASDEX applies the star-coloring recovery described in Section 2.8.2. With n_c colors, the numerical stage requires n_c Hessian-vector products instead of $3N$ coordinate products. Setup, compilation, decompression, and GPU utilization determine the actual wall-time saving. The pattern and coloring are reused for LLPR members at the same geometry.

The sparse Hessian is densified and padding is removed before eigendecomposition; only the AD stage exploits sparsity.

`no_shadow=True` stops adaptive-neighbor-selection derivatives and uses the selected graph. Shadow derivatives introduce extra couplings; `shadow=true` is rejected in this sparse path. Check consistency with MD forces. With three PET blocks, the paper derives $K = 7$, so the configured `hops=3` is an approximation requiring convergence.

Truncation can both discard distant Hessian entries and contaminate retained entries when omitted couplings share a seed color. Enforcing the acoustic sum rule does not remove these errors. The chosen hop count therefore needs comparison with an exact-pattern or feasible dense Hessian, checking signed frequencies and the heat-capacity correction across temperatures. Convergence can be nonmonotonic near mode thresholds; periodic image folding also requires cell/supercell convergence.

Table 4: Numerical controls for Hessian evaluation. Template settings are not established convergence values; archived results use float64.

Control	Effect	Template
`hops`	Changes the assumed sparsity approximation	3
`chunk_size`	Batches colored directions; affects memory/throughput	1
`remat`	Recomputes intermediates to reduce memory	true
`dtype`	Sets floating-point precision	float32

Keep separately tagged convergence outputs, including higher precision and less truncated references, especially for low-frequency modes.

4.5 Normal modes and quantum harmonic heat capacity

Symmetrize the Hessian, enforce the acoustic sum rule, and mass weight:

$$\sum_j H_{i\alpha,j\beta} = 0, \tag{113}$$

The mass-weighted eigenproblem is Eq. (96). Uniform translations should have zero restoring force. Unlike the upstream SADMOF observable, which maps negative eigenvalues to zero, the local calculation preserves signed imaginary frequencies. Strict hybrid assembly rejects unexpected imaginary modes and allows at most three translational near-zero modes. Exploratory outputs instead discard modes at or below the threshold (Section 5.5).

For retained positive modes, use the oscillator heat capacity derived in Section 2.3.3:

$$x_k = \frac{\hbar\omega_k}{k_\mathrm{B}T} = \frac{h_\mathrm{P}c\tilde{\nu}_k}{k_\mathrm{B}T}, \tag{114}$$

$$C_{V,k} = k_\mathrm{B}\left[\frac{x_k/2}{\sinh(x_k/2)}\right]^2, \tag{115}$$

$$c_V(T) = \frac{1}{M_\mathrm{cell}}\sum_{k\in\mathcal{V}} C_{V,k}, \tag{116}$$

$\tilde{\nu}_k$ is wavenumber, and M_cell includes guests. Using cell mass in grams and mode heat capacity in J/K gives J/g/K. Equivalently use $N_\mathrm{A}/M_\mathrm{molar}$. One spectrum supplies the configured 100–500 K curve every 10 K; temperature changes populations, assuming fixed force constants.

4.6 Model uncertainty and propagation

4.6.1 LLPR model uncertainty and CEA propagation

Finite-trajectory sampling error is distinct from potential uncertainty. LLPR varies the final readout of a fixed PET representation [27]. For central energy $\bar{U} = \boldsymbol{w}_0^\mathrm{T}\boldsymbol{f}(\boldsymbol{x})$ and covariance-set features $\boldsymbol{F}$,

$$\begin{aligned}\sigma^2_\mathrm{LLPR}(\boldsymbol{x}) = \alpha^2 \boldsymbol{f}(\boldsymbol{x})^\mathrm{T} \\ \times \left(\boldsymbol{F}^\mathrm{T}\boldsymbol{F} + \lambda\boldsymbol{I}\right)^{-1}\boldsymbol{f}(\boldsymbol{x}).\end{aligned} \tag{117}$$

The regularizer λ stabilizes weakly constrained directions; α is calibrated against energy residuals on a disjoint reference-labeled set using the base potential's electronic-structure definition. Predictions cannot calibrate their own error.

The exported `energy_uncertainty` is an energy standard deviation. Property propagation instead needs persistent, signed energy variations:

$$\begin{aligned}\boldsymbol{w}_i &\sim \mathcal{N}\left[\boldsymbol{w}_0, \alpha^2\left(\boldsymbol{F}^\mathrm{T}\boldsymbol{F} + \lambda\boldsymbol{I}\right)^{-1}\right], \\ U^{(i)}(\boldsymbol{x}) &= \boldsymbol{w}_i^\mathrm{T}\boldsymbol{f}(\boldsymbol{x}),\end{aligned} \tag{118}$$

The exported `energy_ensemble` preserves member identity across configurations. Its large-ensemble spread approaches the analytical LLPR value, but cannot capture bias shared by the frozen representation.

Following Imbalzano *et al.* [29], central-trajectory frames supply

$$\Delta U_t^{(i)} = U_t^{(i)} - \bar{U}_t, \tag{119}$$

$$H_t^{(i)} = K_t + U_t^{(i)} + PV_t, \tag{120}$$

$$\left\langle H^{(i)}\right\rangle_i^\mathrm{direct} = \frac{\sum_t e^{-\beta\Delta U_t^{(i)}} H_t^{(i)}}{\sum_t e^{-\beta\Delta U_t^{(i)}}}, \tag{121}$$

Extensive energy differences can collapse the exponential weights. The primary large-system estimate uses first-order cumulant expansion (CEA):

$$\begin{aligned}\left\langle H^{(i)}\right\rangle_i^\mathrm{CEA} = \left\langle H^{(i)}\right\rangle_{\bar{U}} \\ - \beta\,\mathrm{cov}_{\bar{U}}\left(H^{(i)}, \Delta U^{(i)}\right), \\ \beta = (k_\mathrm{B}T)^{-1}.\end{aligned} \tag{122}$$

Direct reweighting remains a diagnostic: $N_\mathrm{eff} = 1/\sum_t \tilde{w}_t^2$ should remain adequate, and $\mathrm{var}_t(\beta\Delta U_t^{(i)}) \ll 1$ is needed for controlled first-order CEA.

Average replicas member by member, preserve indices across temperatures, and differentiate each complete enthalpy curve:

$$\begin{aligned}C_{P,\mathrm{cl}}^{(i)}(T) &= \frac{\mathrm{d}\langle H^{(i)}\rangle_i^\mathrm{CEA}}{\mathrm{d}T}, \\ \sigma_{\mathrm{LLPR},C_P}(T) &= \left\{\frac{1}{M-1}\sum_{i=1}^{M}\left[C_{P,\mathrm{cl}}^{(i)}(T) - \overline{C}_{P,\mathrm{cl}}(T)\right]^2\right\}^{1/2}.\end{aligned} \tag{123}$$

The final property spread is a predictive standard deviation; do not divide it by $\sqrt{M}$.

4.6.2 Gaussian NPT propagation

Direct propagation of shallow ensembles (DPOSE) combines member properties before measuring their spread [28]. The saved comparison uses a Gaussian NPT approximation on central-model trajectories. Define $Q_i = U_i + PV$ and $\Delta U_i = U_i - U_0$. Joint Gaussianity leaves covariance unchanged under exponential reweighting, giving

$$C_{P,i}^{\mathrm{G}} \simeq \frac{f}{2} k_{\mathrm{B}} + \frac{\mathrm{var}_0(Q_i)}{k_{\mathrm{B}} T^2}. \tag{124}$$

Here $f = 3N - 3$. The central-model fluctuation estimator is exact under the usual equilibrium assumptions; using central-trajectory variances for other members needs Gaussian closure. Pair member indices with harmonic corrections.

The main Gaussian plots replace only the propagated model spread around the central enthalpy-derivative curve. The separate variance-based hybrid also replaces the central classical value and estimates sampling error from squared configurational-enthalpy fluctuations. Neither guarantees smaller or better calibrated uncertainty.

4.7 Optimized minima and the hybrid correction

Relax the equilibrated empty structure directly at fixed cell for a separate reference. For loaded systems, quench representative independent replicas with the same MLIP and accept only force-converged, stable minima. Raw thermal frames are not harmonic references. Current outputs use one geometry per model/loading from 400 K.

For N_m accepted minima,

$$\overline{\Delta c^{\mathrm{har}}}(T) = \frac{1}{N_m} \sum_{s=1}^{N_m} \Big[c_{\mathrm{qn},s}^{\mathrm{har}}(T) - c_{\mathrm{cl},s}^{\mathrm{har}} \Big], \tag{125}$$

$$s_{\Delta c}^2(T) = \frac{1}{N_m - 1} \sum_{s=1}^{N_m} \Big[\Delta c_s^{\mathrm{har}}(T) - \overline{\Delta c^{\mathrm{har}}}(T) \Big]^2. \tag{126}$$

Use the same retained modes for quantum and classical terms, with k_{B} per classical mode. Mixed host–guest motion requires the complete loaded spectrum.

$$\begin{aligned} c_P^{\mathrm{approx}}(T) &= c_{P,\mathrm{cl}}^{\mathrm{anh}}(T) + \overline{\Delta c^{\mathrm{har}}}(T), \\ c_{P,\mathrm{cl}}^{\mathrm{anh}}(T) &= \frac{1}{M_{\mathrm{cell}}} \frac{\mathrm{d}\langle H \rangle_{\mathrm{cl}}}{\mathrm{d}T}. \end{aligned} \tag{127}$$

This combines an anharmonic classical C_P with a harmonic C_V correction, assuming a small quantum pressure–volume term. Validation below about 200 K remains limited.

Persistent LLPR members must contribute both terms before their spread is taken:

$$\sigma_{\mathrm{model}}^2 = \sigma_{\mathrm{cl,model}}^2 + \sigma_{\mathrm{har,model}}^2 + 2\,\mathrm{cov}_i(c_{P,\mathrm{cl}}^{(i)}, \Delta c_{\mathrm{har}}^{(i)}), \tag{128}$$

$$u_{\mathrm{hyb}}^2 = \sigma_{\mathrm{MD}}^2 + \sigma_{\mathrm{har,samp}}^2 + \sigma_{\mathrm{model}}^2. \tag{129}$$

The covariance is across matched model members. σ_{MD} is the sampling error of the central enthalpy derivative; $\sigma_{\mathrm{har,samp}}$ comes from independently quenched structures. Quadrature assumes independence of the sampling and model contributions.

Current single-replica analysis perturbs enthalpy means with Gaussian draws using autocorrelation-based errors and recomputes the derivative. It supplies within-trajectory error, not a block bootstrap or between-replica variance. Bands omit the unestimated harmonic sampling term: $u_{\mathrm{hyb}} = (\sigma_{\mathrm{MD}}^2 + \sigma_{\mathrm{model}}^2)^{1/2}$. LLPR Hessians use central-model geometries without member-specific relaxation.

5 Results and discussion

5.1 Available calculations and status

The campaign contains 66 trajectories: PET-MAD (`pet-mad-1.5-s-40nn`) and PET-SOL (`pet-sol-s-best`), three loadings, eleven temperatures, and one replica per state point. Each 500 ps run contributes 8001 frames after discarding 100 ps: 33 ns simulated and 26.4 ns analyzed in total. Hybrid curves span 200–400 K; 175/425 K support centered endpoint derivatives. Each curve reuses one loaded 400 K reference spectrum.

All estimates are *exploratory*: every loaded spectrum has imaginary modes, and four of six relaxations failed their convergence criterion. Discarding unstable modes does not satisfy the strict acceptance protocol. Bands quantify selected sampling and model errors, excluding unstable-mode bias and unestimated variation across minima.

Trajectory diagnostics come from per-run `summary.json`; central values and relaxation diagnostics come from `heat-capacity.exploratory-discard-imaginary` CSV/JSON archives. Reported bands use refreshed stride-1 evaluation of all 8001 frames with 64 persistent LLPR members, rather than older per-loading uncertainty columns. Figure sources are recorded in `results-provenance.json`.

5.2 Thermodynamic sampling and equilibration

Mean temperatures stay within 0.315 K of target. PET-MAD/100 at 300 K gives 299.854 ± 8.031 K (mean and standard deviation), close to the equipartition width $T\sqrt{2/(3N-3)} = 8.058$ K. This checks kinetic sampling, not configurational ergodicity. Pressure fluctuations span 0.80–1.51 $\times 10^3$ bar with mean errors of 9–18 bar; PET-MAD/100 gives 4.65 ± 13.11 bar, compatible with the 1 bar target. Physical enthalpy always uses imposed pressure.

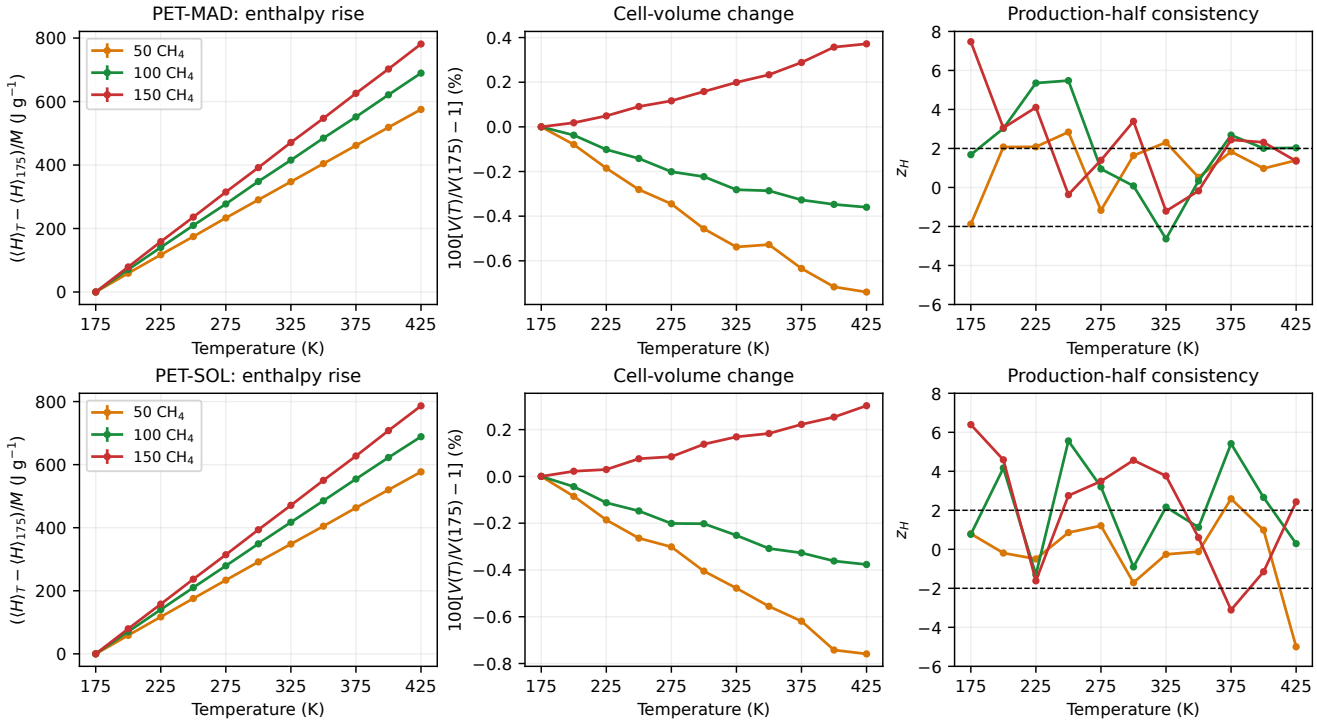


Figure 9: Thermodynamic diagnostics from all 66 trajectory summaries. Left: mass-specific enthalpy increase relative to the same model and loading at 175 K; the baseline is a display offset, and the small bars show the pointwise enthalpy standard errors without propagating the shared baseline error. Middle: fractional change of mean cell volume relative to 175 K. Right: the archived difference between production-half mean enthalpies divided by their combined standard error, z_H; dashed lines mark ± 2. These are diagnostics of the individual runs, not replica averages.

Enthalpy increases smoothly (Figure 9). Archived enthalpy correlation times of 0.095–0.101 ps imply 1980–2115 effective samples and mean errors of 0.015–0.046 eV. These short-time estimates do not exclude slow drift.

Half-trajectory discrepancies flag 33 of 66 runs with $|z_H| > 2$, where

$$z_H = \frac{\overline{H}_{\text{second}} - \overline{H}_{\text{first}}}{\sqrt{\text{SE}^2_{\text{first}} + \text{SE}^2_{\text{second}}}}. \tag{130}$$

Examples are $z_H = 7.47$ for PET-MAD/150 at 175 K and 4.57 for PET-SOL/150 at 300 K. Some RMS-force correlation times reach 26.1 ps, leaving fewer than eight effective samples.

In Figure 10, the accumulated mean differs from the final mean by about −0.30 eV after 100 ps and −0.15 eV after 200 ps. A large model spread does not absorb a temperature-dependent sampling bias. Longer independent runs and sensitivity to discarded intervals are needed to validate the reported small MD errors.

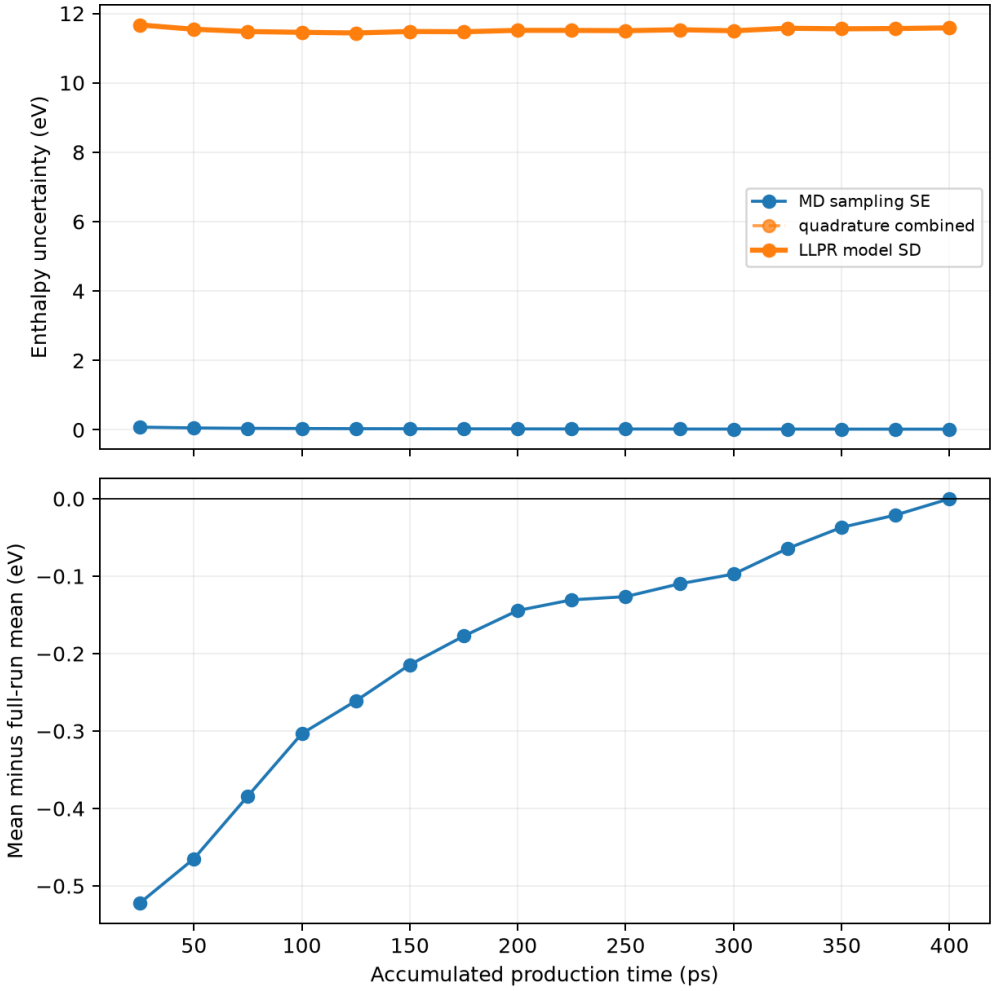


Figure 10: Archived enthalpy convergence diagnostic for PET-MAD with 150 methane molecules at 175 K. The upper panel separates MD sampling error from the LLPR spread of the *enthalpy*, in eV. The lower panel shows the accumulated mean relative to the full-production mean. Its endpoint is zero by construction and is not evidence of equilibration.

5.3 Cell response, framework motion, and methane mobility

Both models predict contraction at 50/100 guests and expansion at 150 over 175–425 K (Figure 9). PET-MAD volumes change from 17146.8 to 17019.9, 17170.2 to 17108.4, and 17210.4 to 17 274.4 Å^3, respectively. Across models the changes are about −0.75%, −0.37%, and +0.3–0.4%. Densities at 300 K are about 0.677, 0.752, and 0.825 g/cm^3, largely reflecting added guest mass. Because $C_P - C_V \propto \alpha_V^2$, contraction does not imply a negative heat-capacity difference; compressibility is also needed to quantify it.

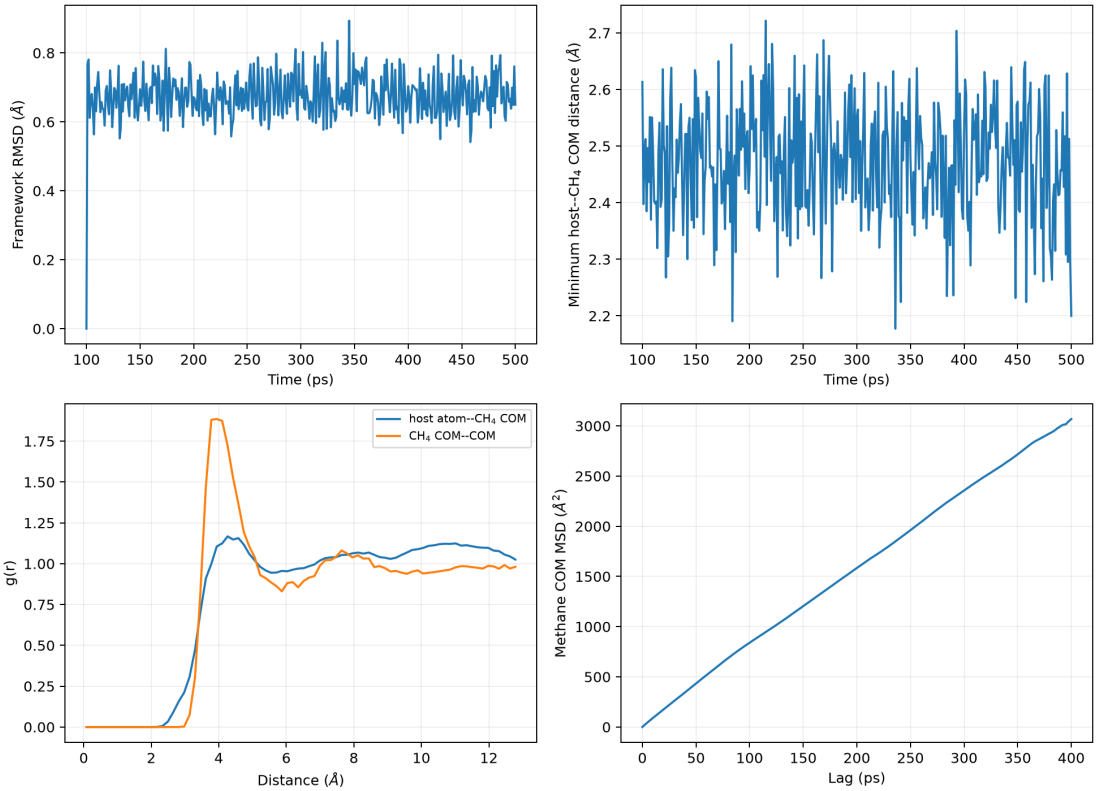


Figure 11: Structural diagnostics for PET-SOL with 150 methane molecules at 300 K: framework RMSD, nearest host-atom–methane center-of-mass distance, radial distributions, and methane center-of-mass mean-squared displacement (MSD). The first analyzed frame at 100 ps defines the framework reference, hence the initial zero RMSD. Long-lag MSD values use fewer time origins than short-lag values.

At 300 K, framework RMSDs are 0.659–0.725 Å, while tracked bond-length changes are only 0.055–0.059 Å. Bounded RMSD supports collective motion without gross collapse over the analyzed interval, but tracking a reference neighbor set cannot rule out all bond changes.

Methane 400 ps MSDs at 300 K decrease with loading: 8.9–9.2, 5.1–5.6, and 3.1–3.4 × 10^3 Å^2 for 50, 100, and 150 molecules, respectively. Guest radial distributions peak near 4 Å. Mobility and crowding support the need for anharmonic MD; a single harmonic geometry cannot describe diffusion. Endpoint MSDs alone do not establish diffusion coefficients without a verified diffusive regime, cell-motion control, and replica statistics.

5.4 Classical, harmonic, and hybrid heat capacities

Table 5 resolves the 300 K contributions. The negative harmonic correction removes 59–63% of the classical term, as quantum occupations suppress high-frequency vibrations. Its increase in magnitude with loading reflects added methane and mixed host–guest modes, requiring loaded rather than empty spectra.

Room-temperature hybrid values rise from approximately 0.93 to 1.07–1.08 and 1.16 J/g/K with loading. Model differences of 0.003, 0.011, and 0.001 J/g/K are below propagated bands and cannot distinguish predictive accuracy.

The curves generally rise over 200–400 K (Figures 12–13). For PET-MAD/50, the classical term decreases from 2.347 to 2.269 J/g/K, while the correction becomes less negative, from −1.551 to −1.164 J g^{-1} K^{-1}. Thus the upward hybrid slope need not indicate increasing classical anharmonicity. Adjacent loading bands overlap; significance requires paired propagation of their difference. The grid does not bracket the reference low-temperature minimum, and small local bends remain unresolved within uncertainty.

Empty-framework 300 K values are 0.786 ± 0.021 (PET-MAD) and 0.766 ± 0.027 J g^{-1} K^{-1} (PET-SOL). These are framework-mass-normalized quantum harmonic c_V values with LLPR spread, not hybrid c_P or incremental adsorption heat capacities.

Table 5: Heat capacities and uncertainty estimates at 300 K, all in J/g/K and normalized by the total mass of each loaded cell. c_{cl} is the central enthalpy derivative, Δc_{har} the retained-mode harmonic correction, and c_{hyb} their sum. u_{CEA} and u_{G} use different paired model spreads around the *same* central value and MD standard error σ_{MD}. The final column is the separate variance-based central hybrid estimate. None of these uncertainties includes the bias from discarded imaginary modes or an estimated spread across minima.

Model	$N_{\mathrm{CH_4}}$	c_{cl}	Δc_{har}	c_{hyb}	σ_{MD}	u_{CEA}	u_{G}	$c_{\mathrm{hyb}}^{\mathrm{var}}$
PET-MAD	50	2.281	-1.351	0.929	0.010	0.068	0.127	0.926
PET-MAD	100	2.762	-1.682	1.080	0.010	0.123	0.169	1.044
PET-MAD	150	3.122	-1.957	1.165	0.010	0.113	0.214	1.202
PET-SOL	50	2.289	-1.363	0.926	0.010	0.087	0.150	0.940
PET-SOL	100	2.762	-1.693	1.069	0.010	0.153	0.225	1.024
PET-SOL	150	3.134	-1.969	1.166	0.010	0.103	0.306	1.230

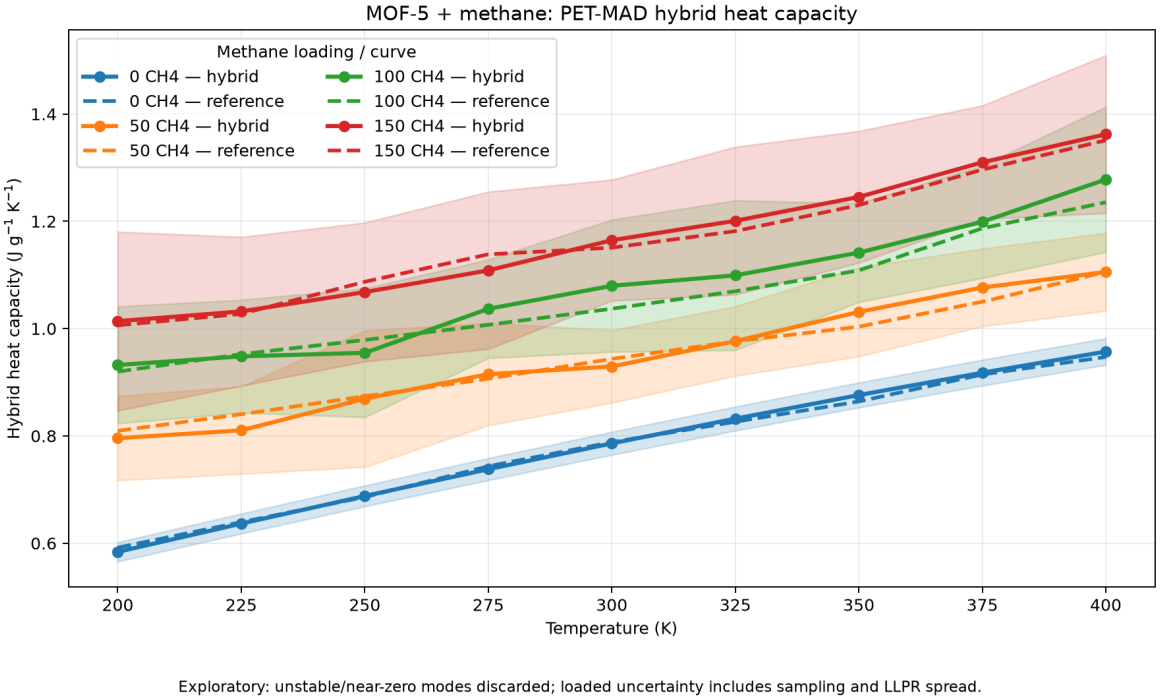


Figure 12: Supplied PET-MAD loading comparison. Loaded solid curves are central classical enthalpy derivatives plus retained-mode harmonic quantum corrections. Their shading is $\pm u_{\mathrm{CEA}}$, combining MD sampling error with correlated classical–harmonic LLPR spread. Despite the common "hybrid" legend, the zero-loading curve is a quantum-harmonic c_V calculation only, with a harmonic LLPR band; no empty-system MD term was computed. Dashed curves are the supplied reference CSVs, whose provenance is discussed in the text.

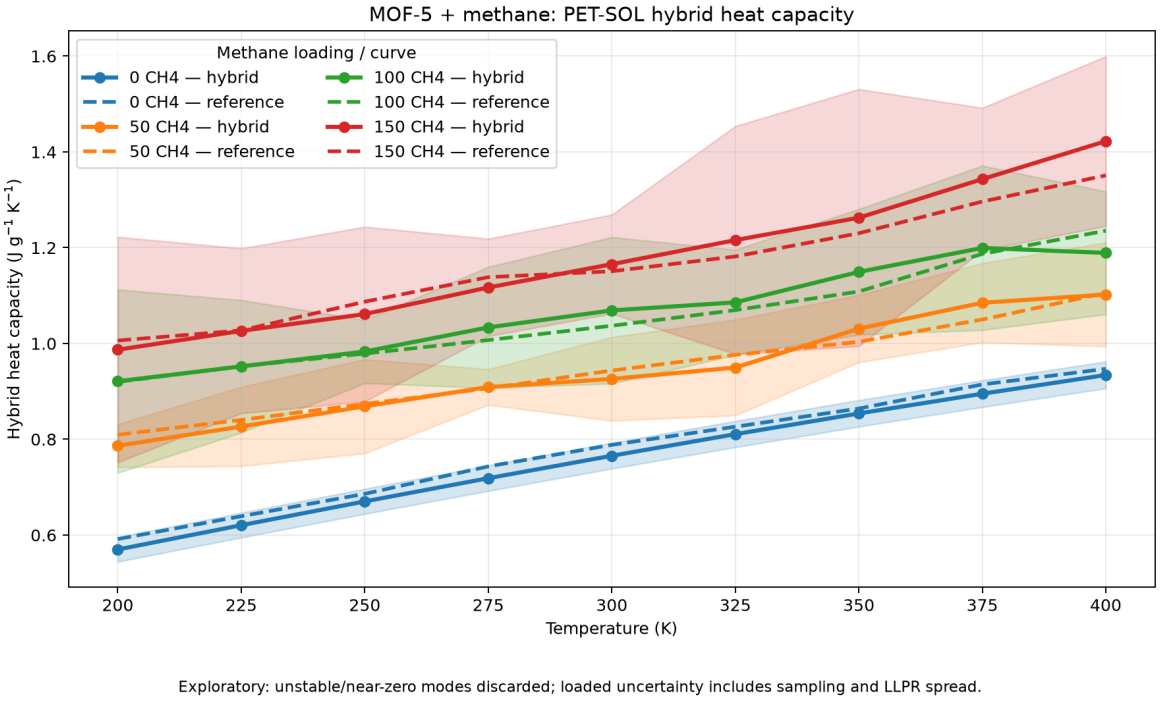


Figure 13: Supplied PET-SOL loading comparison, with the same definitions and exploratory mode treatment as Figure 12. The slight decrease of the 100-molecule central curve at the upper end of the grid is smaller than its uncertainty band and is not evidence of a resolved thermodynamic anomaly. Shading denotes standard uncertainty, not a calibrated confidence interval.

Dashed curves are digitized Kapil *et al.* data [21]. Their 300 K values are approximately 0.788, 0.944, 1.037, and 1.151 J/g/K for 0/50/100/150 guests. Loaded differences from this work are at most 0.043 J/g/K. Agreement is contextual: the digitization has no uncertainty estimates, the force fields and preparation differ, and present sampling/spectra remain unconverged. The earlier approximate empty-host scale is not a separate fitting target.

5.5 Hessian stability

Table 6: Archived central loaded-Hessian diagnostics. All calculations use PET-JAX, float64, three graph hops, and the force-constant row-sum acoustic treatment. "Converged" reproduces the relaxation archive's flag for its requested tolerance. Imaginary modes have signed frequencies below $-1\,\mathrm{cm}^{-1}$; near-zero modes satisfy $|\widetilde{\nu}| \leq 1\,\mathrm{cm}^{-1}$. Retained modes have positive frequencies above the threshold. Signed negative values represent unstable curvature, not physical negative-frequency oscillators.

Model	$N_{\mathrm{CH_4}}$	Converged	$F_{\max}$ (eV/Å)	Imaginary	Near zero	Retained	$\widetilde{\nu}_{\min}$ (cm^{-1})
PET-MAD	50	no	0.0170	17	5	2000	-53.5
PET-MAD	100	yes	0.0097	114	3	2655	-92.4
PET-MAD	150	no	0.0095	28	3	3491	-66.8
PET-SOL	50	no	0.0054	32	3	1987	-107.7
PET-SOL	100	yes	0.0100	95	3	2674	-138.4
PET-SOL	150	no	0.0049	35	3	3484	-86.5

Force convergence does not establish positive curvature: the two 100-molecule structures pass their archived force criteria but retain 114/95 imaginary modes, down to $-92.4/-138.4\,\mathrm{cm}^{-1}$. Other loaded runs have 17–35 imaginary modes and failed convergence flags; PET-MAD/50 also has five near-zero modes. Run-dependent criteria must be judged from each archive rather than comparing residual forces alone. Empty spectra have four/five imaginary modes, down to $-5.5/-6.6\,\mathrm{cm}^{-1}$.

Discarding negative eigenvalues enables an oscillator sum but does not create a stable minimum. Distinguish insufficient relaxation, unstable stationary structures, and sparse-curvature errors with convergence checks. Matching quantum/classical mode masks avoids inconsistent subtraction; it does not estimate unstable-mode bias. Reusing one 400 K geometry also leaves temperature dependence and minimum-to-minimum variation untested. LLPR spread at a central geometry does not replace structural sampling.

5.6 Uncertainty bands

Figures 14–15 separate MD sampling, classical LLPR, harmonic LLPR, and paired total uncertainty. MD errors are about 0.010 J/g/K at 300 K; model terms dominate. Covariance can make the total smaller than an individual component (Eq. (128)).

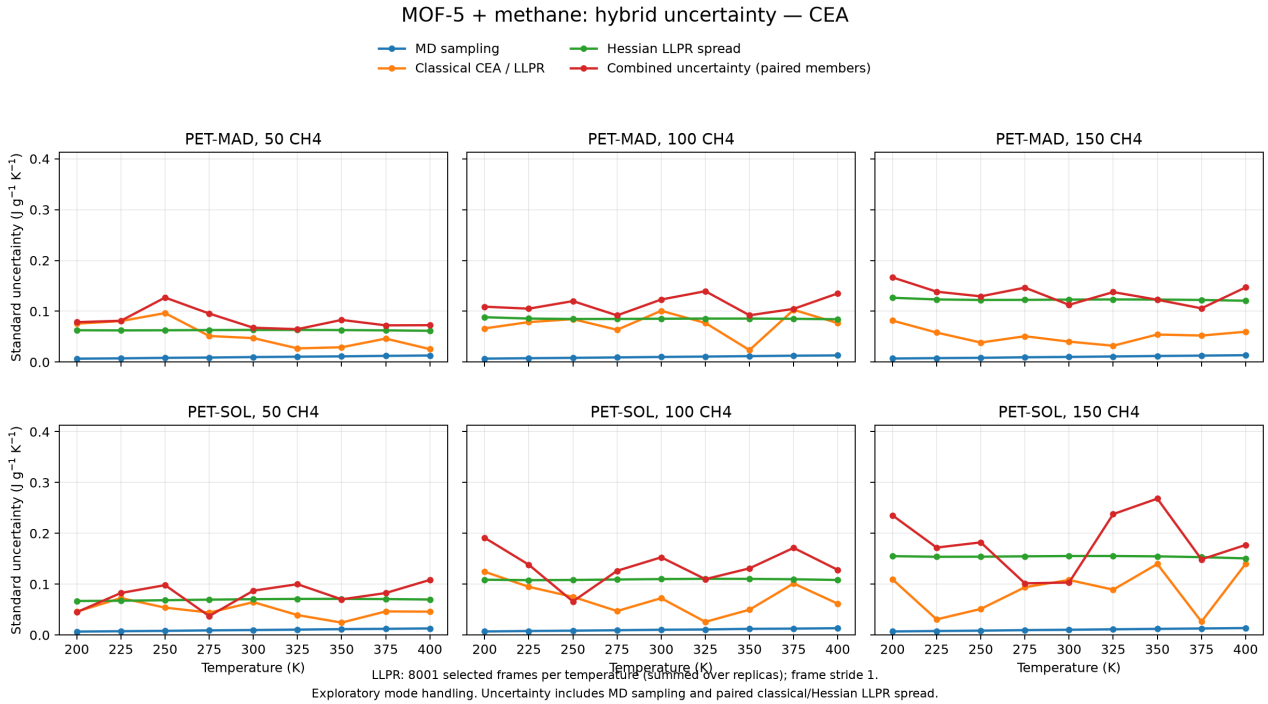


Figure 14: Supplied uncertainty decomposition using refreshed CEA member curves. Every panel uses all 8001 saved production frames at each temperature and 64 LLPR members. The harmonic curve is model spread at one fixed geometry, not an uncertainty across minima. The total preserves classical–harmonic covariance and excludes an unestimated harmonic sampling term. All plotted quantities are standard errors or model standard deviations in J/g/K, not variance fractions.

For PET-SOL with 150 guests at 300 K, classical and harmonic CEA spreads of 0.108 and 0.155 J/g/K combine to 0.103 J/g/K, with inferred correlation −0.75. PET-SOL/100 instead has correlation +0.37 and paired spread 0.153 J/g/K. Independent quadrature would miss both effects; predictive spread is not reduced by $\sqrt{64}$.

Gaussian propagation gives larger 300 K bands in every case (Table 5), with positive classical–harmonic correlations of 0.43–0.63. Smoother or narrower bands do not demonstrate calibration. Recorded $\mathrm{var}(\beta\Delta U_i)$ spans about 2.4 to 1.42×10^3, violating the small-variance CEA criterion; some direct reweighting effective counts approach one. Neither collapsed reweighting nor untested Gaussian tails validate these bands, whose coverage against reference electronic-structure data remains unestablished.

The separate `variance-hybrid` calculation also changes the central classical estimator. At 300 K, hybrid differences from the derivative route range from -0.045 to $+0.064\,\mathrm{J\,g^{-1}\,K^{-1}}$, with MD errors of 0.022–0.030 J/g/K from squared-fluctuation correlations. Shared trajectories and harmonic corrections make these cross-checks correlated. Quantitative predictions require stable Hessians, stationary independent trajectories, multiple quenched structures, and validated member propagation.

6 Recommended production and analysis protocol

Validate timestep through short NVE energy-drift tests before reservoir coupling, which can mask integration errors. Cell relaxation requires a stress-capable, validated model. One-cell Hessians and selected minima need finite-size and configurational convergence; their unweighted spread is not automatically a free-energy-weighted average. Transfer of the reference hybrid approximation to PET, especially below 200 K, needs validation.

7 Implementation, outputs, and reproducibility

`mof_heat_capacity.protocols.loaded` prepares campaigns; `scripts/md/submit_loaded_md.sh` submits loaded NPT and rejects zero-methane MD. Property stages use the scripts under `scripts/properties/`: `submit_analysis.sh` for diagnostics, `submit_heat_capacity.sh` for fixed-cell relaxation and PET-JAX Hessians, and

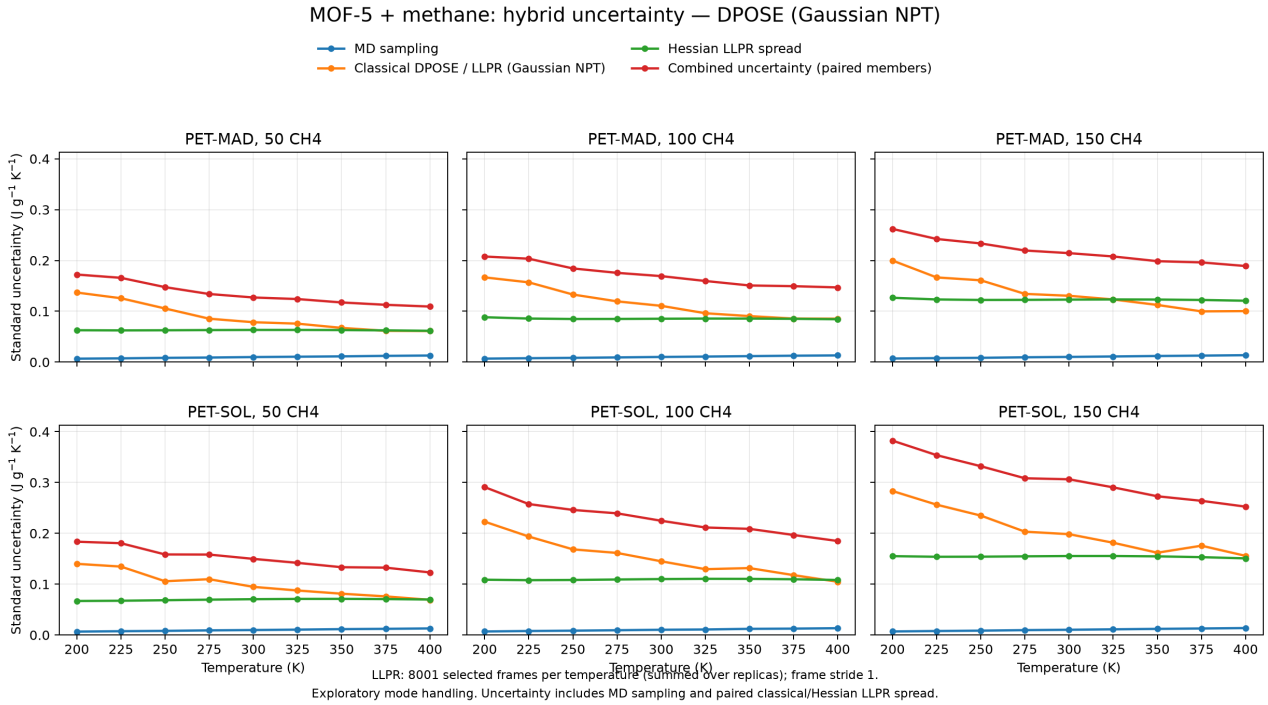


Figure 15: Supplied Gaussian NPT/DPOSE uncertainty decomposition. Central hybrid curves, MD sampling errors, harmonic member corrections, and frame coverage are identical to the CEA comparison; only the classical member spread and its covariance with the harmonic correction change. The smoother orange curves result from a different estimator and do not establish more accurate uncertainty calibration.

Table 7: Production requirements for the workflow in Figure 1. Reference agreement supplements internal convergence; it is not a fitting target.

Calculation	Acceptance and convergence requirements
Empty reference	Relax the equilibrated host directly at fixed cell with the selected checkpoint. Report forces and signed modes; retain a separately normalized harmonic curve.
Loaded NPT	Independent placement/velocity seeds. Check stationary enthalpy, cell, bonds, pores, and guest motion; converge duration, replicas, cell size, timestep, coupling times, and temperature spacing.
Loaded minima	Quench representative independent configurations at fixed cell. Reject unconverged or unstable structures.
Hessians	Converge hops, precision, HVP batching, acoustic treatment, frequency threshold, and supercell/phonon sampling on the correction.
Hybrid assembly	Average accepted loaded corrections; differentiate replica enthalpies; pair LLPR indices. Report MD, minimum sampling, and model uncertainties separately and jointly.

`submit_hybrid_analysis.sh` for final assembly. Hessian LLPR members run as eight GPU array tasks of eight members each; `analysis.llpr_merge` supplies the final hybrid dependency.

Archives retain structures, signed frequencies, model/run identities, grids, precision, hops, chunking, and rematerialization. Hybrid NPZ/CSV/JSON outputs store the classical term, harmonic correction, sum, uncertainties, and provenance.

Table 8: Uncertainty components, their calculation locations, and their stored outputs.

Quantity	Where it is calculated	Stored result
LLPR energy standard deviation	metatrain LLPR covariance and calibration	`energy_uncertainty`; copied as `analytical_energy_uncertainty_eV` in the per-run archive
CEA member enthalpies and overlap diagnostics	`analysis/uncertainty.py`, function `committee_enthalpy_estimates`	per-run `model_uncertainty.npz`
Classical LLPR C_P standard deviation	`analysis/results.py`, after member-wise replica averaging and temperature differentiation	`model_uncertainty_heat_capacity.npz` and `.csv`
MD sampling standard error	`analysis/statistics.py`, then replica combination and derivative resampling in `analysis/hybrid.py`	`classical_anharmonic_cp_standard_error`
Harmonic sampling standard error	`analysis/hybrid.py`, across independent optimized minima	`harmonic_quantum_correction_standard_error`
Hessian LLPR standard deviation	`analysis/harmonic.py`, after differentiating persistent LLPR readouts at the central minimum	`harmonic_quantum_correction_model_standard_deviation`
Combined standard uncertainty	paired model variance and sampling quadrature in `analysis/hybrid.py`	`approximate_cp_combined_standard_uncertainty`

The scalar energy uncertainty is diagnostic; property errors require complete persistent member curves. Paired MD/harmonic readouts preserve covariance, but Hessians use central-model minima without member-specific relaxation.

Use one node, one task, and one GPU per MD/Hessian job; parallelize independent states, replicas, minima, or member tasks. Retain Slurm IDs and measure representative runtime/memory before adjusting requests:

```
sacct -j JOB_ID --format=JobID,State,Elapsed,MaxRSS,AllocTRES
```

Use distinct outputs for every loading, temperature, seed, and replica.

Record structure/composition and insertion criteria; checkpoint, export, conversion, and environment identities; LLPR dataset/ensemble hashes, calibration, regularizer, seeds, member count, stride, and overlap diagnostics; ensemble/integration settings and restart history; relaxation forces, signed modes, and Hessian controls; and production intervals, replica counts, correlation/block methods, and convergence comparisons.

8 Limitations and interpretation

Path integral molecular dynamics could test the neglected quantum anharmonic contribution, particularly for water-loaded frameworks. It samples quantum nuclear statistics through an auxiliary ring polymer, but is not used for the results reported here. Such validation would require convergence in bead number and trajectory sampling [4, 16].

9 Conclusions

PET-MAD and PET-SOL give exploratory room-temperature heat capacities rising from about 0.93 to 1.16 J/g/K with methane loading. Harmonic quantization removes 59–63% of the classical term; guest mobility supports retaining anharmonic MD. Broad agreement with digitized reference trends does not establish quantitative accuracy. Stable spectra, independent trajectory/minimum sampling, and validated paired uncertainty propagation are needed before interpreting these curves as converged predictions.

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Table 9: Limits of the present estimates and the evidence needed to resolve them.

Present limitation	Required evidence
33/66 enthalpy drift flags; one replica per state	Longer independent trajectories and stable block/replica statistics.
Imaginary modes in every loaded reference; four failed relaxations	Stable minima and converged signed spectra, sparsity reach, and precision.
One loaded geometry per curve	Multiple independent quenches and sensitivity to harmonic-reference temperature.
Classical MD plus harmonic C_V correction	Validation of quantum anharmonic and pressure–volume effects, especially below 200 K.
Fixed-loading random insertion	Demonstrated equilibration of guest distributions; separate particle-exchange sampling for adsorption equilibrium.
Shallow LLPR; large CEA diagnostics and method-dependent bands	Reference-data calibration and validated propagation; shared representation bias and member-specific relaxation remain outside the current spread.
PET differs from QuickFF; finite-cell/numerical approximations	Loaded-MOF energy/force/stress/curvature benchmarks and size/grid convergence.

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